

Quantifying comprehensive marine vessel emissions using data fusion

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Last Updated on September 10, 2026

Table of contents

1	Double counting between the AIS-based and S1-unmatched estimates	3
2	In-line manuscript statistics	3
2.1	Abstract	3
2.2	Introduction	3
2.3	Results	3
2.4	Discussion	5
2.5	Methods	5
3	Results	7
3.1	Figure 1	7
3.2	Figure 2	8
3.3	Figure 3	10
3.4	Figure 4	11
4	Methods	12
4.1	Figure 6	12
4.2	Table 1	12
4.3	Double counting between the AIS-based and S1-unmatched estimates	12
4.4	Double counting tables	14
4.5	Figure 7	15
4.6	Table 2	17
4.7	Figure 8	17
4.8	Figure 9	18
4.9	Figure 10	19
4.10	Figure 11	20
4.11	Table 3	21

4.12	Figure 12	23
4.13	Figure 13	24
4.14	Figure 14	25
4.15	Figure 15	26
4.16	Figure 16	27
4.17	Table 4	29
4.18	Table 5	29
4.19	Figure 17	29
4.20	Figure 18	31
5	Supplementary materials: Supplementary figures	33
5.1	Figure S1	33
5.2	Figure S2	34
5.3	Figure S3	35
5.4	Figure S4	37
5.5	Figure S5	38
6	Supplementary materials: Supplementary tables	39
6.1	Table S1	39
6.2	Table S2	39
6.3	Table S3	40
6.4	Table S4	40
6.5	Table S5	40
7	Supplementary materials: Supplementary figures: Comparing our emission estimates to other inventories	41
7.1	Figure S6	41
7.2	Table S6	43
7.3	Table S7	43
7.4	Figure S7	43
7.5	Figure S8	44
7.6	Figure S9	45
8	Inventories comparison	46
8.1	Inventory intensity across models	58

1 Double counting between the AIS-based and S1-unmatched estimates

2 In-line manuscript statistics

2.1 Abstract

- We found that vessels emitted approximately 1.53 billion metric tonnes of CO₂ in 2025. In 2024, the most recent year for which EDGAR data are available, marine vessels accounted for 16% of global fossil fuel emissions from the transportation sector, and 3.6% of global fossil fuel emissions across all sectors.
- Current AIS-only approaches underestimate the magnitude of 2025 emissions by 22%. Meanwhile, because AIS coverage has generally been increasing over time, current AIS-only approaches overestimate the increasing emissions trend by 15% (from 2017 to 2025).

2.2 Introduction

- Climate change mitigation requires accurate accounting across sectors; while the transportation sector as a whole was responsible for 21% of all global CO₂ emissions in 2024, fundamental gaps remain in understanding the contribution by marine vessels.
- For over 775,000 AIS-broadcasting vessels, we quantify emissions for each vessel using precise latitude/longitude/timestamp information for over 119 billion individual AIS messages.
- To account for the vast missing emissions from non-broadcasting vessels (also sometimes known as ‘dark vessels’), we develop a machine learning framework that leverages 31.9 million vessel detections from across more than 924,000 S1 scenes to estimate non-broadcasting emissions at a monthly global 1x1-degree gridded resolution, from 2017 through 2025, and disaggregated by vessel type (fishing or non-fishing).

2.3 Results

- Using this data fusion approach, we find that total global CO₂ emissions at sea in 2025 was 1.53 billion MT, with an increase of 38% from 2017 to 2025. If only considering emissions from vessels that broadcast AIS, in line with currently accepted state-of-the-art models, global CO₂ emissions in 2025 would be only 1.19 billion MT, with a more pronounced increase from 2017 to 2025 of 44%.

- Thus using only AIS data from broadcasting vessels underestimates the magnitude of 2025 global CO₂ emissions by 22%, and overestimates the increasing trend for 2017–2025 by 15%
- In 2025, 36% of CO₂ emissions from fishing vessels were from non-broadcasting vessels, and 21% of CO₂ emissions from non-fishing vessels were from non-broadcasting vessels. For both types of vessels, this fraction of emissions from non-broadcasting vessels has generally been declining since 2021, although it increased slightly for fishing vessels in 2025.
- Across the entire 2017 to 2025 time series, almost all non-broadcasting emissions (96.4%) occurred inside the S1 footprint and in pixel-months that were imaged
- Our machine learning model for predicting these emissions directly from observed S1 detection density achieves a high out-of-sample predictive performance (traditional R² = 0.94 and total percent error of -8.6% for CO₂)
- Only 3.5% of non-broadcasting emissions occurred inside the S1 footprint but in pixel-months that weren't imaged by S1, requiring us to use a two-stage hurdle framework using both a classification model (F1 Score = 0.72) and a regression model (traditional R² = 0.73) to predict S1 detection density before predicting emissions (traditional R² = 0.94).
- Only 0.079% of non-broadcasting emissions occurred outside the S1 footprint entirely, since most vessels operating in these open ocean areas use AIS.
- China, Japan, and United States had the most CO₂ emissions during within-country domestic trips in 2025.
- China, Malaysia, and United States had the most CO₂ emissions during international trips that started or ended in these countries in 2025 (where the emissions from each trip are partitioned equally to its departure and arrival country).
- China, United States, and Indonesia had the most CO₂ emissions during port stays.
- Of the roughly 775,000 AIS-broadcasting vessels in our analysis, we matched 139,000 (18%) to public registries or proprietary databases, the majority of which (93,400) were IMO registered.
- Because many large vessels are already listed with the IMO, the majority of our total CO₂ emissions come from these vessels (39%); 19% come from vessels listed on other registries; and 42% come from vessels with no public registry information and which rely entirely on modeled vessel characteristics.
- Although vessel activity in the Southern Ocean represents a very small fraction of the world's marine vessel emissions (0.04%), it represents the part of the world experiencing the highest growth rate of marine vessel emissions (267%).

- These polar estimates are primarily driven by AIS-broadcasting vessels, with only a small fraction coming from non-broadcasting vessels (1.02% in the Southern Ocean).
- Polar regions have extremely limited S1 coverage (only 0.06% of pixel-months in the Southern Ocean were imaged by S1), meaning that trends there may be more susceptible to AIS adoption biases.
- Using the latest data available from the EU’s Emissions Database for Global Atmospheric Research (EDGAR), we estimate that in 2024 marine vessels accounted for 16% of all transportation-related CO₂ emissions. Marine vessel CO₂ emissions increased by 32% between 2017 and 2024, 10 times faster than the growth rate of other transportation emissions, and 5 faster than the growth rate of total global emissions across all sectors.

2.4 Discussion

- For non-fishing vessels, the total social cost of CO₂, CH₄, and N₂O emissions in 2021 was 344 billion USD. This represents about 5% of the total value of all goods transported by maritime trade (using 6.63 trillion USD as the 2021 value of maritime-traded goods).
- For fishing vessels, the total social cost of CO₂, CH₄, and N₂O emissions in 2022 was 24 billion USD. This represents about 15% of the total value of all wild capture fisheries (using 159 billion USD as the 2022 value of wild capture fisheries).

2.5 Methods

- The time between pings is typically small (the mean value is 0.0881 hours and the median value is 0.0447 hours, or about 5 minutes and 3 minutes respectively), while the maximum value for this across our dataset is 47 hours (meaning this is the most amount of time over which we will estimate emissions for a single message).
- Double counting between the two estimates. The detection-AIS matching considers AIS positions up to 12 hours either side of a detection, but a hold-out experiment by GFW (detections with high-confidence matches as ground truth, AIS positions withheld within a window of the detection) shows that the probability of missing a true match rises gradually with the length of the gap in a vessel’s AIS messages rather than switching on at 12 hours: 6.8% of true matches are missed at a 2-hour gap, 27.4% at 8 hours and 45.4% at 18 hours. Weighting the share of AIS-based CO₂ emitted in gaps of each length by that miss rate, 5.1% of AIS-based CO₂ (4.5% if the miss rate is held at its 18-hour value beyond the last simulated gap) is emitted while the vessel, had it been imaged, would have been classed as unmatched. This implies that 0.92 million of the 14.9 million unmatched detections (6.2%) are AIS-broadcasting vessels, and hence that about 6.2% of S1-unmatched CO₂ (175 million tonnes over 2017-2025, or 1.5% of the combined AIS-based + S1-unmatched total) duplicates emissions already counted in the

AIS-based estimate. Because the same misclassified detections are also missing from the matched densities the emissions regression is trained on, the fitted emissions per detection are high by a factor of 1.054; folding that in gives an upper end of 11.9% of S1-unmatched CO₂ (2.9% of the combined total). This replaces the earlier reasoning that only messages representing more than 12 hours could be affected, which happened to give a similar number (5.0%) for the wrong reason: most of the exposure comes from gaps of 2 to 18 hours (Fig. ??, Table 2, Table 3).

- In all, we analyze AIS data for over 770,000 AIS-broadcasting vessels, across 119 billion individual timestamped AIS messages which are assigned to 14.4 million international trips, 112 million domestic trips, and 96.4 million port visits that occurred between 2017 and 2025.
- As a result, we used a dataset containing 31.9 million vessel detections from 2017 to 2025, of which 14.9 million detections were not matched to a broadcasting vessel ('unmatched'). These detections come from across more than 924,000 individual S1 scenes.
- We were able to use this direct estimation approach for 96.4% of the total estimated non-broadcasting CO₂ emissions .
- Of the total predicted non-broadcasting CO₂ emissions, 3.5% corresponded to indirect estimations within the S1 footprint, while only 0.079% occurred entirely outside the footprint
- For a validation sample of 16,205 vessels, updating vessel metadata with our random forest models substantially improves main engine model performance: traditional R² rises from 0.73 using main engine power and design speed from the IMO lookup table, to 0.85 using our random forest predictions, approaching the performance ceiling of 0.9 achieved with the metadata recorded in the registered dataset.
- Validating against the EU MRV database, performance peaks at a traditional R² of 0.84 for vessels whose times at sea in our dataset are within 15% of the MRV-reported values (validation sample of 27,657 vessel-year observations). Beyond this threshold, performance declines as discrepancies grow between our annual aggregated trip activity and that recorded in the MRV dataset.

3 Results

3.1 Figure 1

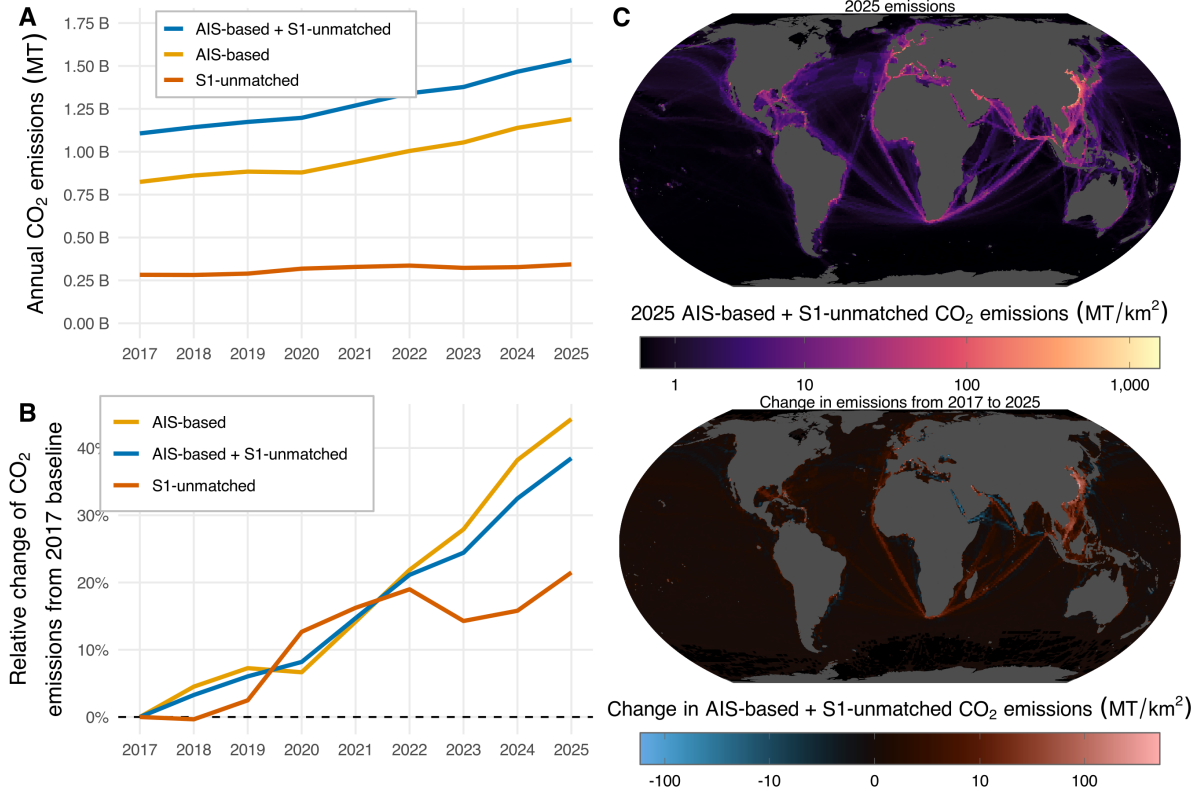


Figure 1: (A) Annual marine vessel CO₂ emissions, with each line colored based on the data source. (B) Relative change in annual marine vessel CO₂ emissions from a 2017 baseline, with each line colored based on the data source. (C) Spatial map of CO₂ emissions in 2025 (top) and change in CO₂ emissions between 2017 and 2025 (bottom), aggregated across AIS-broadcasting vessels and non-broadcasting vessels, and shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

3.2 Figure 2

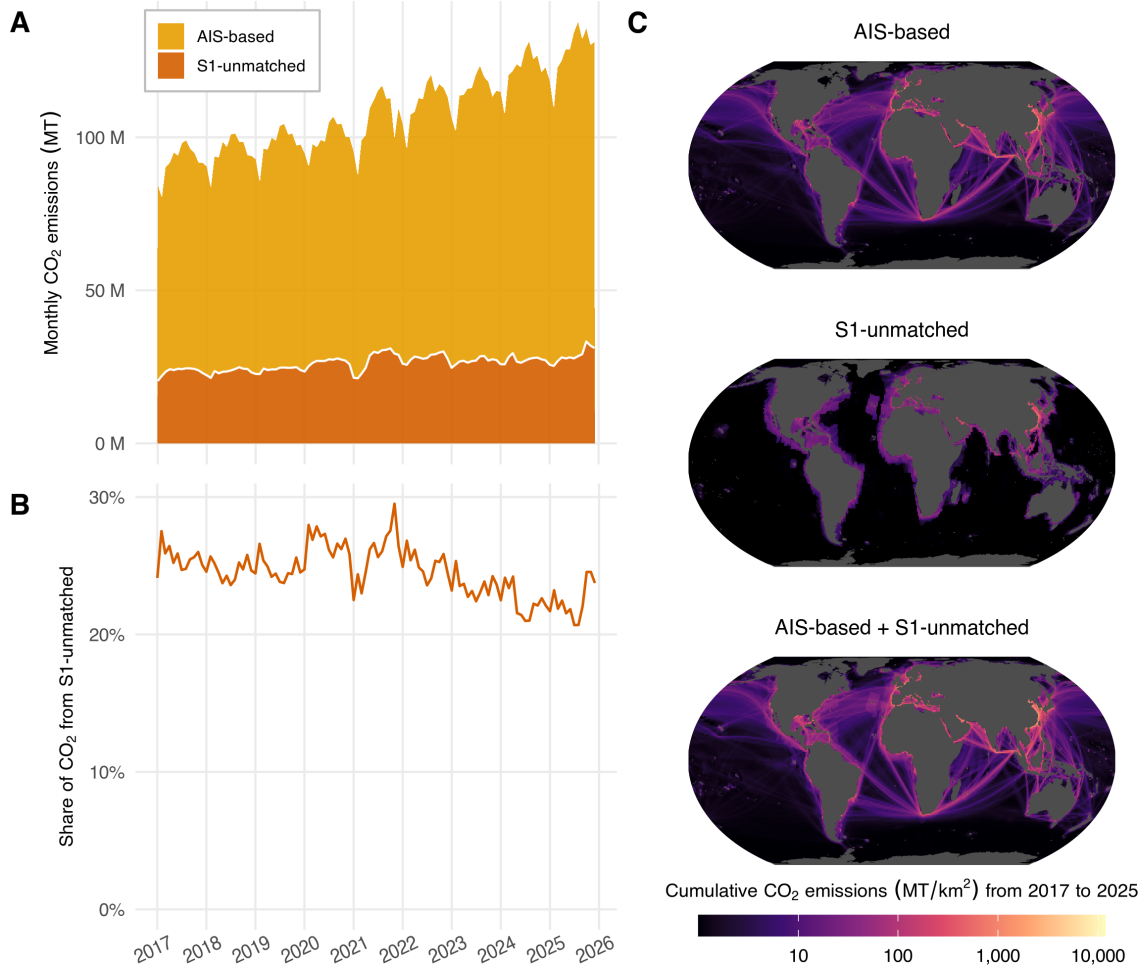


Figure 2: (A) Time series of monthly global CO₂ emissions, shown for both AIS-broadcasting and S1-detected non-broadcasting vessels. (B) Time series of the monthly share of global CO₂ emissions attributable to S1-detected non-broadcasting vessels. (C) Spatial distribution of cumulative 2017-2025 CO₂ emissions from AIS-broadcasting vessels (top), S1-detected non-broadcasting vessels (middle), and both fleets combined (bottom), shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

3.3 Figure 3

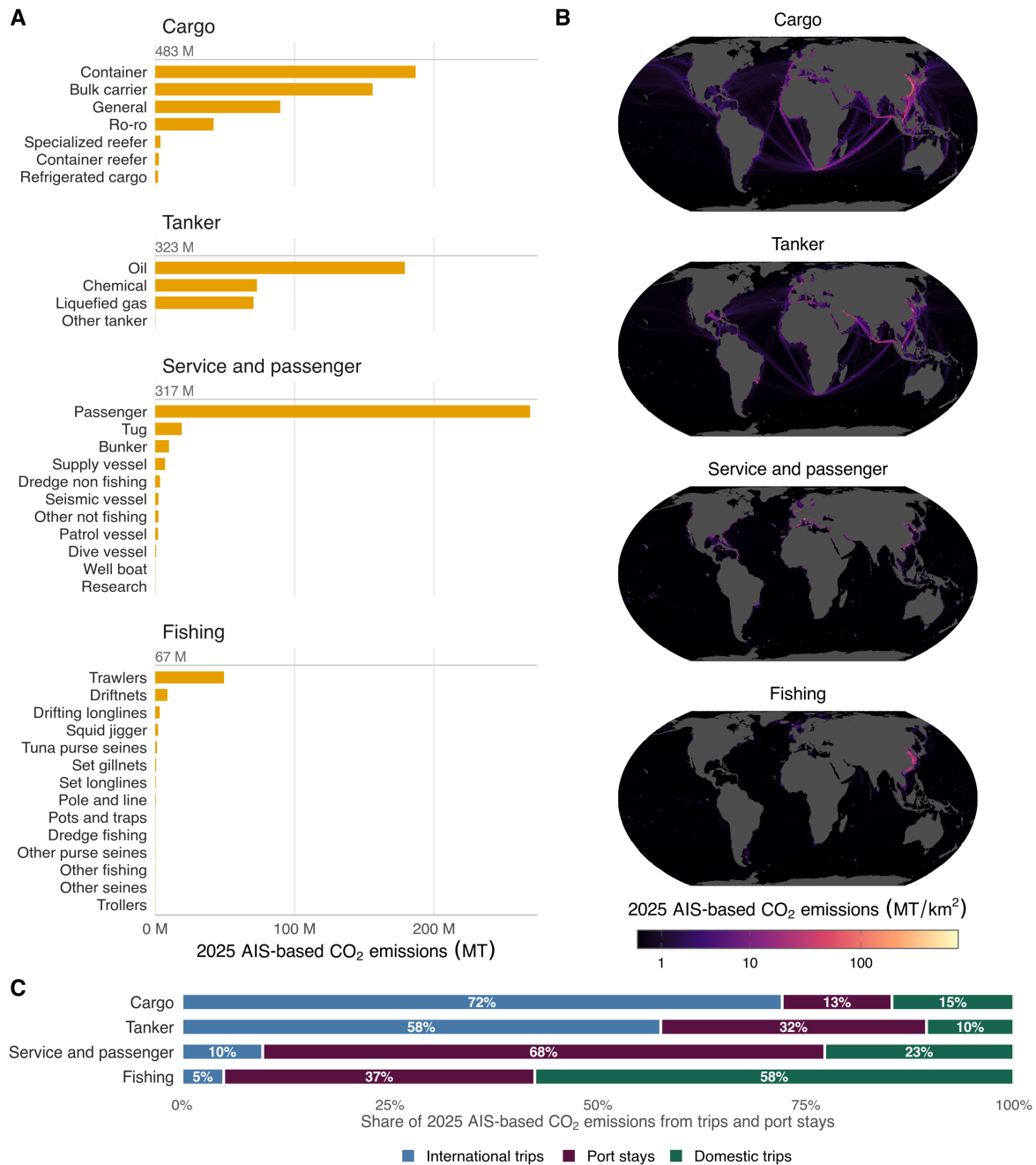


Figure 3: Richness of AIS-broadcasting vessel emissions data, showing: (A) Total 2025 CO₂ emissions by vessel class for AIS-broadcasting vessels. Vessel classes are grouped into four families, with the total for each family shown above its bars; every class in a family is shown as its own bar, so panel heights differ with the number of classes. These totals are aggregated from the same gridded emissions shown in panel B, so the two panels describe exactly the same emissions and the bars sum to the mapped total. (B) Spatial distribution of 2025 CO₂ emissions from AIS-broadcasting vessels for each of those same vessel class families, shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale). Families appear in the same order in both panels. (C) For each family, the share of CO₂ emissions attributable to international trips, port stays, and domestic trips (note that Panel C is not a decomposition of panel A: it covers only emissions that could be assigned to a trip or port visit that ended in 2025; each family sums to 100% of its own assigned emissions)

3.4 Figure 4

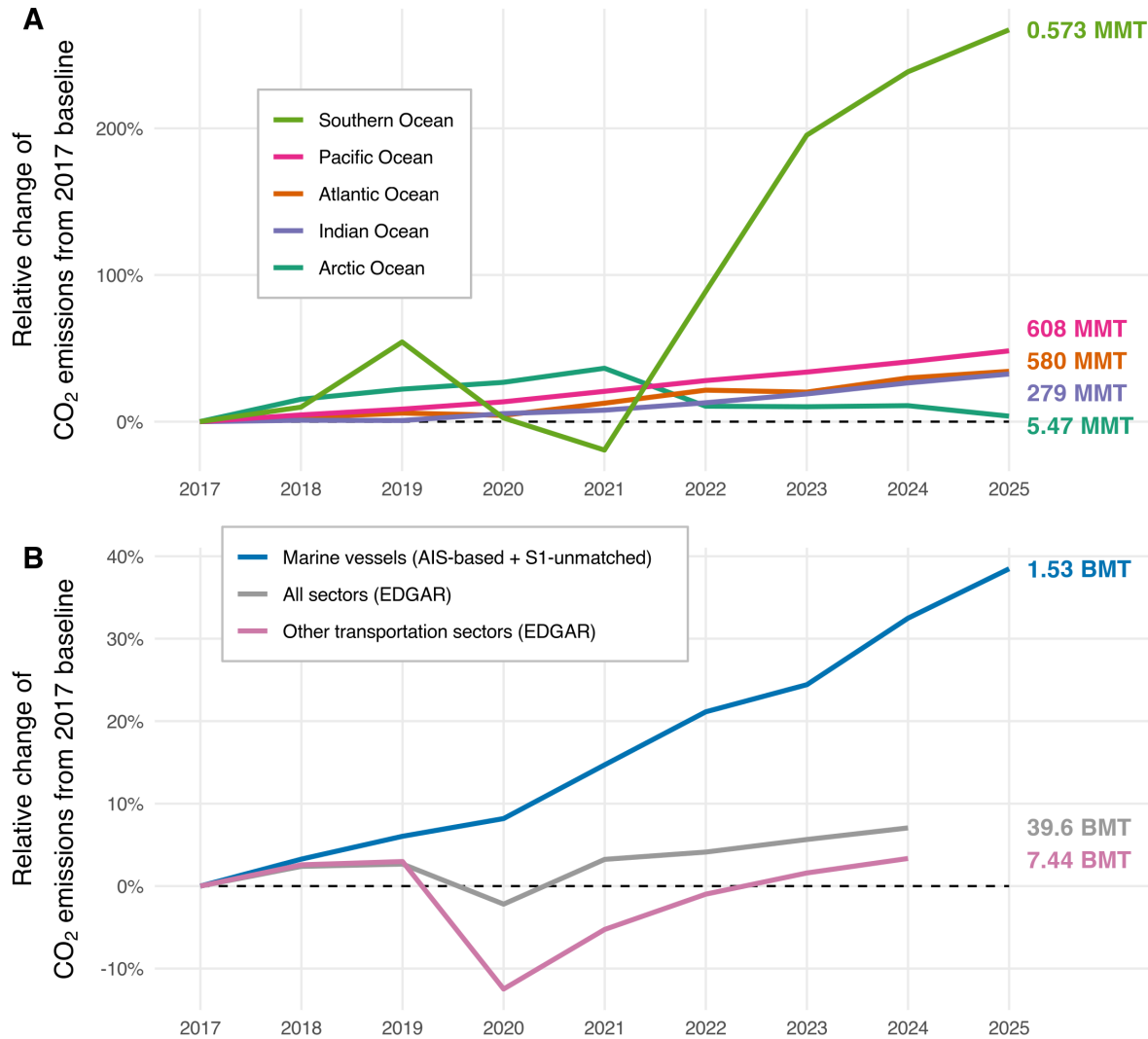


Figure 4: (A) Trend in CO₂ emissions by ocean (normalized to 2017 baseline), using our fused AIS+S1 marine vessel data. (B) Comparison in trend of CO₂ emissions data (normalized to 2017 baseline) between marine transportation emissions from GFW, other transportation emissions from EDGAR, and all sector emissions from EDGAR. On the right-hand side of the figure, absolute values are shown as text and color-coded for each line, for the last year of available data for that series (2025 for the GFW marine vessel series in panel A and B; 2024 for the EDGAR series in panel B).

4 Methods

4.1 Figure 6

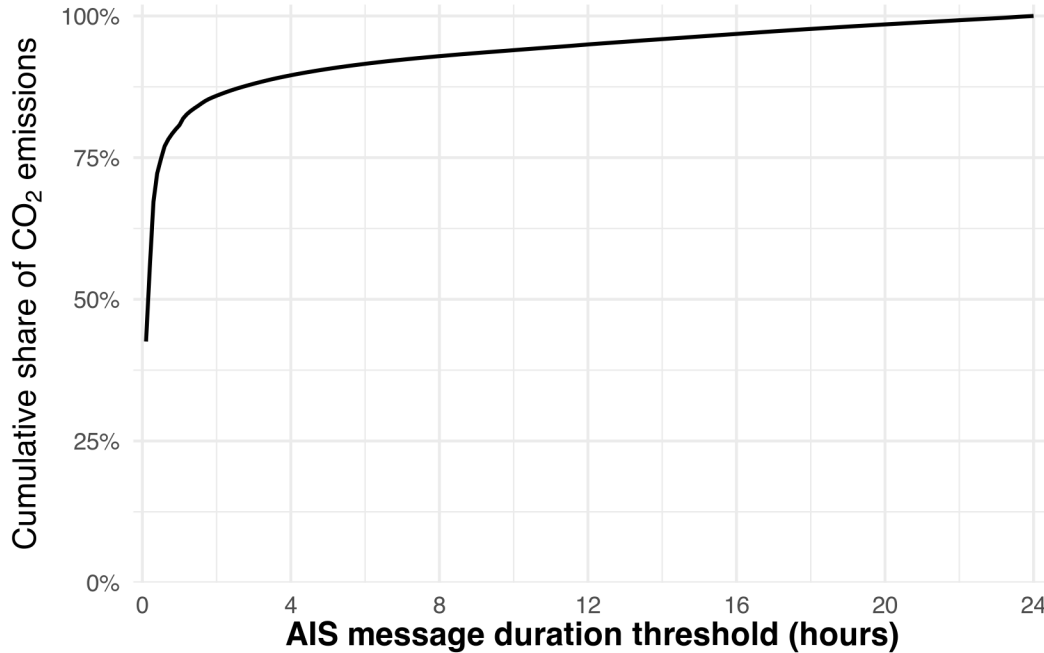


Figure 5: Cumulative share of AIS-based CO₂ emissions attributable to AIS messages whose assigned duration falls at or below a given threshold. Each message is assigned the time elapsed since the previous message from that vessel and emissions are estimated over that duration, so long durations are those over which activity is interpolated across a gap in the message stream rather than observed directly. Values at whole-hour thresholds are tabulated in Table 1.

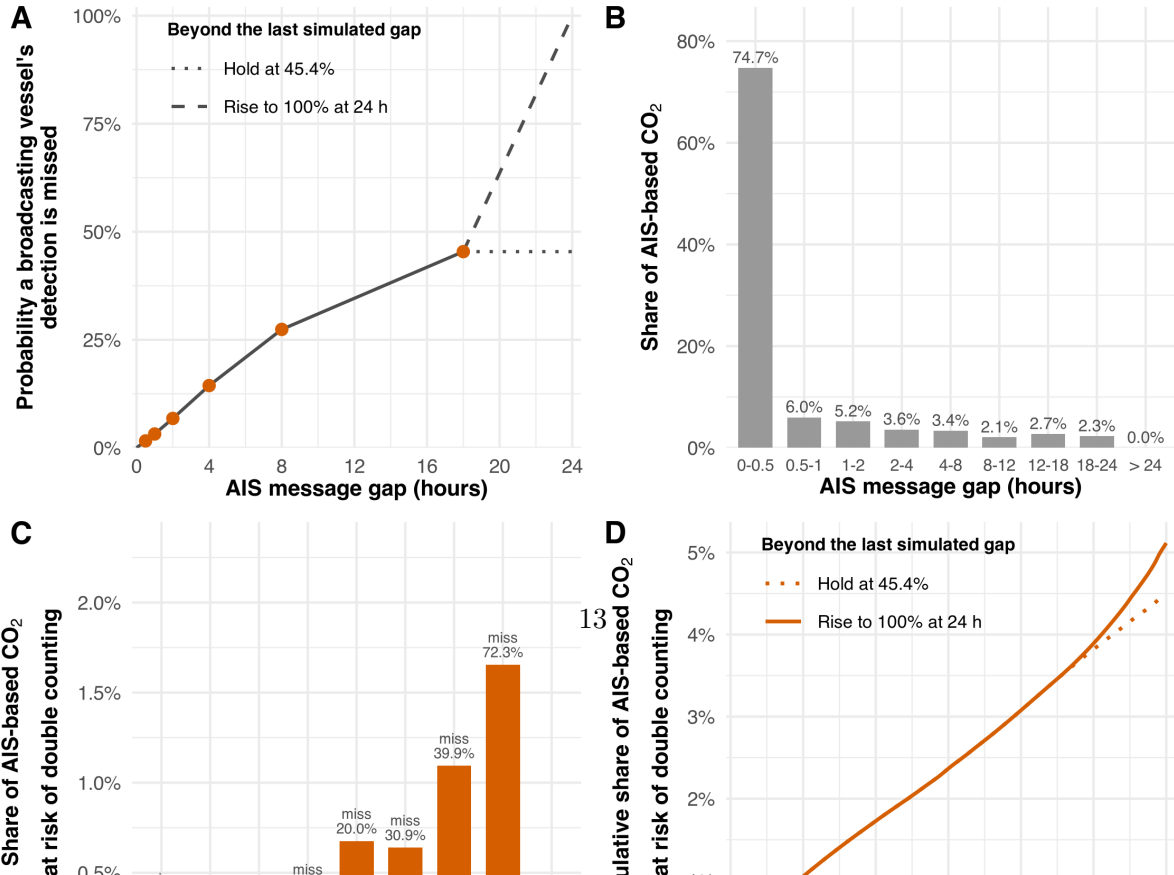
4.2 Table 1

4.3 Double counting between the AIS-based and S1-unmatched estimates

The computation lives in the `s1-matching-double-counting` chunk above the in-line statistics; these chunks only draw and tabulate it.

Table 1: Cumulative AIS-based CO₂ emissions from messages whose assigned duration falls at or below each whole-hour threshold, 2017 to 2025. Shares are of all AIS-based CO₂ emissions, including that from messages with durations longer than 24 hours, so the at-or-below column need not reach 100%. The above-threshold column is its complement: the share of emissions interpolated across gaps longer than the threshold. Figure ?? shows the full curve at 0.1-hour resolution.

Threshold (hours)	Cumulative CO ₂ (million tonnes)	Share at or below	Share above
1	7,086	80.73%	19.27%
2	7,542	85.92%	14.08%
3	7,727	88.04%	11.96%
4	7,859	89.53%	10.47%
5	7,958	90.66%	9.34%
6	8,036	91.55%	8.45%
7	8,100	92.29%	7.71%
8	8,155	92.91%	7.09%
9	8,203	93.46%	6.54%
10	8,247	93.96%	6.04%
11	8,291	94.46%	5.54%
12	8,336	94.97%	5.03%
13	8,378	95.45%	4.55%
14	8,419	95.92%	4.08%
15	8,459	96.37%	3.63%
16	8,499	96.83%	3.17%
17	8,539	97.28%	2.72%
18	8,576	97.71%	2.29%
19	8,612	98.12%	1.88%
20	8,646	98.51%	1.49%
21	8,679	98.88%	1.12%
22	8,711	99.25%	0.75%
23	8,743	99.61%	0.39%
24	8,777	100.00%	0.00%



4.4 Double counting tables

Table 2: AIS-based CO₂ emissions by the length of the AIS gap during which they were emitted, 2017 to 2025, and the share at risk of double counting: the emissions in each row multiplied by the probability that an S1 detection made during a gap of that length is missed by the matching (Fig. ??A). Beyond 18 hours the miss rate is taken to reach 100% at 24 hours; messages representing more than 24 hours are treated as always missed.

Gap (hours)	Share of CO ₂	Miss rate	At risk	Cumulative at risk
0-0.5	74.73%	0.4%	0.32%	0.32%
0.5-1	6.00%	2.2%	0.13%	0.45%
1-2	5.19%	4.6%	0.24%	0.69%
2-4	3.61%	10.2%	0.37%	1.05%
4-8	3.37%	20.0%	0.68%	1.73%
8-12	2.07%	30.9%	0.64%	2.37%
12-18	2.74%	39.9%	1.09%	3.46%
18-24	2.29%	72.3%	1.65%	5.12%
> 24	0.00%	100.0%	0.00%	5.12%

Table 3: Double counting between the AIS-based and non-broadcasting CO₂ estimates, 2017 to 2025, following the steps in the text. The share at risk is the total of Table 2; because S1 images a vessel at an arbitrary moment, it is also the probability that a detection of a broadcasting vessel is missed by the matching. The matched detections are the ones that were not missed, which gives the number missed and their share of the unmatched detections. Non-broadcasting emissions are estimated from unmatched detections, so that share of them duplicates AIS-based emissions. The last two rows add the effect of the missed detections being absent from the matched detections the emissions regression model is trained on. The miss rate beyond 18 hours is taken to reach 100% at 24 hours (Fig. ??A, dashed).

	Value
Share of AIS-based CO ₂ at risk of double counting: emitted during AIS gaps in which a detection of the vessel would be missed	5.1%
Matched S1 detections	17.0 million
Detections of broadcasting vessels missed by the matching	0.92 million
Unmatched S1 detections	14.9 million
Share of unmatched detections that are broadcasting vessels	6.2%
Non-broadcasting CO ₂ that duplicates AIS-based CO ₂ , 2017–2025	175 million tonnes (1.5% of combined total)
Increase in the emissions per detection fitted by the emissions regression model, because the missed detections are absent from its training data	5.4%
Upper bound on non-broadcasting CO ₂ that duplicates AIS-based CO ₂ , including that increase	11.9% of non-broadcasting CO ₂ (2.9% of combined total)

4.5 Figure 7

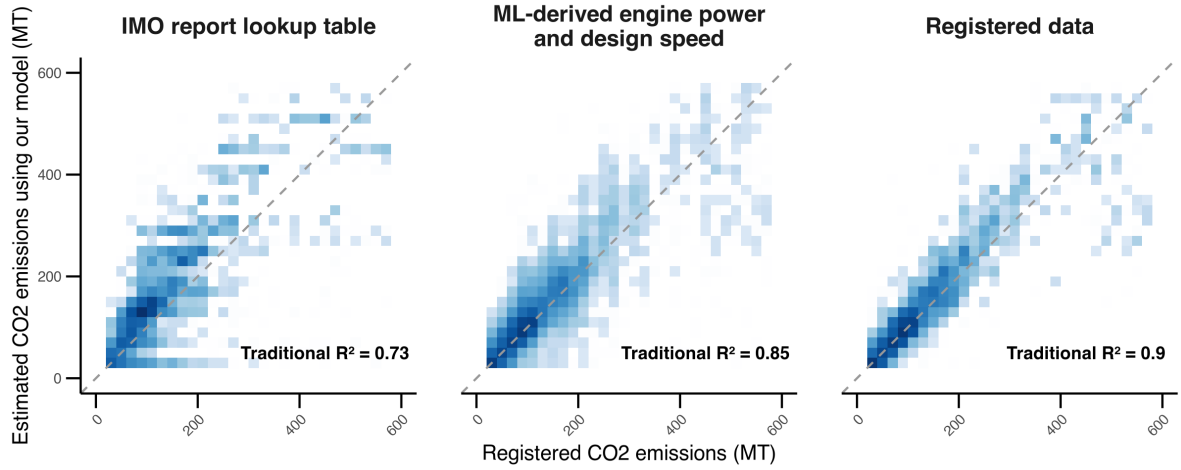


Figure 7: Comparison of daily vessel-level CO_2 emissions estimated using our model with registered emissions from a proprietary vessel database (based on main engine fuel consumption over 24 hours). Estimates are shown using main engine power and design speed from IMO (left), our machine learning random forest models (center), and the same registered dataset (right). The dashed line represents the 1:1 line. Performance values for traditional R^2 are shown for each set of vessel characteristics. Over 16,000 vessels are represented in the figure, and blue shading indicates point density. All data are rule-of-three compliant, meaning each cell (20x20 mt) displayed contains at least three vessels.

Table 4: Performance summary across different thresholds for the hours at sea tolerance between datasets. Results are shown using the R^2 (measuring linear association between observed and predicted values) and traditional R^2 (proportion of variance explained compared to the mean-only baseline).

Tolerance	Number of observations	R^2	Traditional R^2
0.05	11,621	0.84	0.83
0.10	20,432	0.85	0.84
0.15	27,657	0.85	0.84
0.20	33,753	0.85	0.83
0.25	38,980	0.84	0.82
0.30	43,958	0.83	0.81
0.35	48,399	0.83	0.80
0.40	52,490	0.82	0.79

4.6 Table 2

4.7 Figure 8

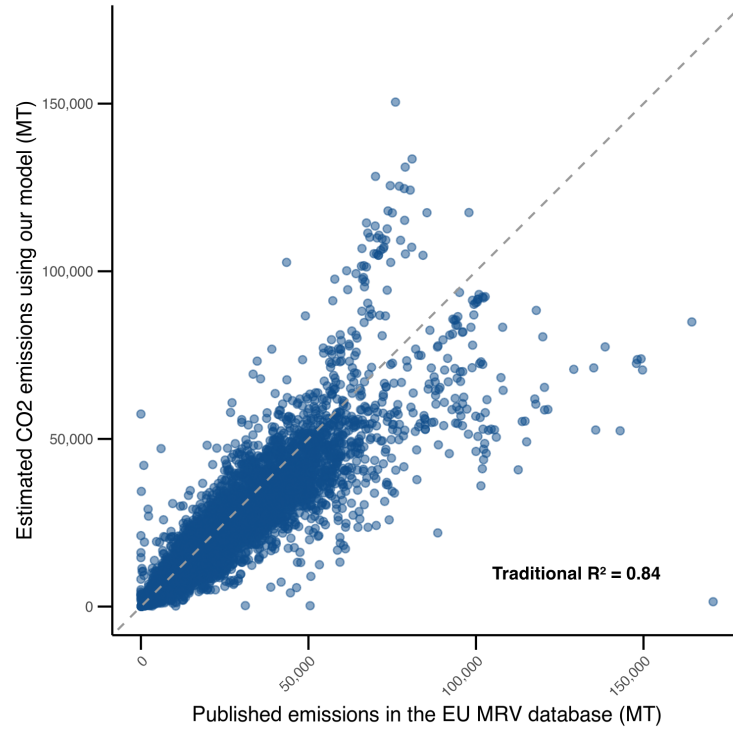


Figure 8: Comparison of annual vessel-level CO₂ emissions estimated using our model with published emissions from the EU MRV database. The sample includes 27,657 observations from yearly aggregated vessel trips where hours at sea differ by no more than 15% between our data and the MRV dataset. The dashed line represents the 1:1 line.

4.8 Figure 9

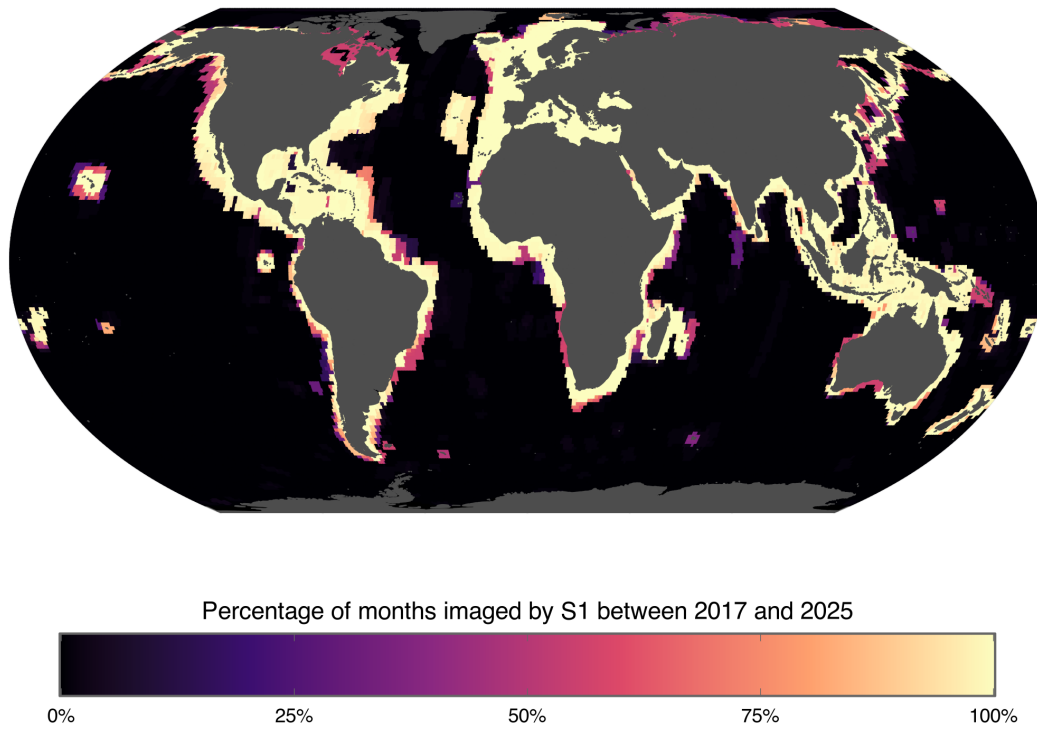


Figure 9: Map showing the percentage of months from 2017 through 2025 that were imaged by S1 for each 1x1 degree pixel.

4.9 Figure 10

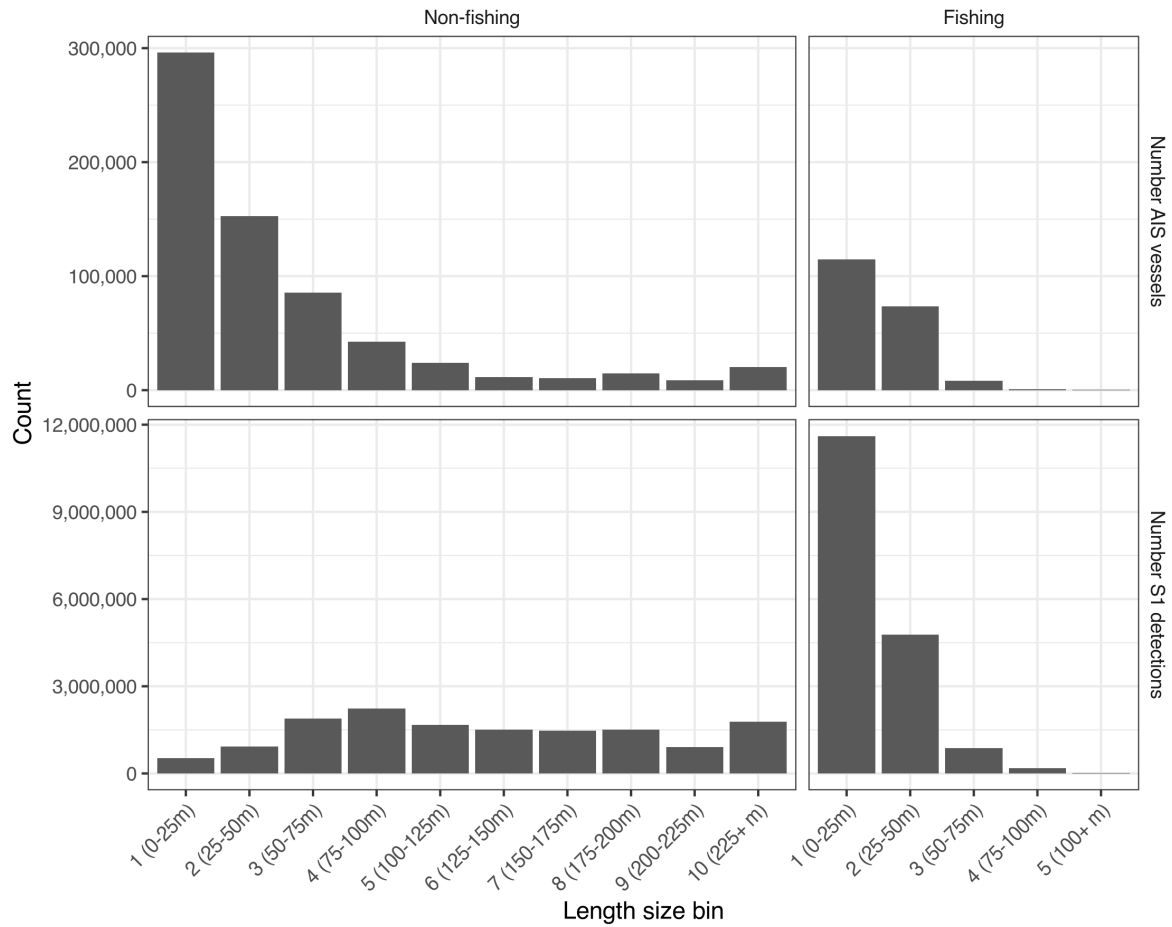


Figure 10: Distributions showing the number of unique AIS vessels and number of S1 detections for each length size bin, disaggregated by fishing and non-fishing vessels.

4.10 Figure 11

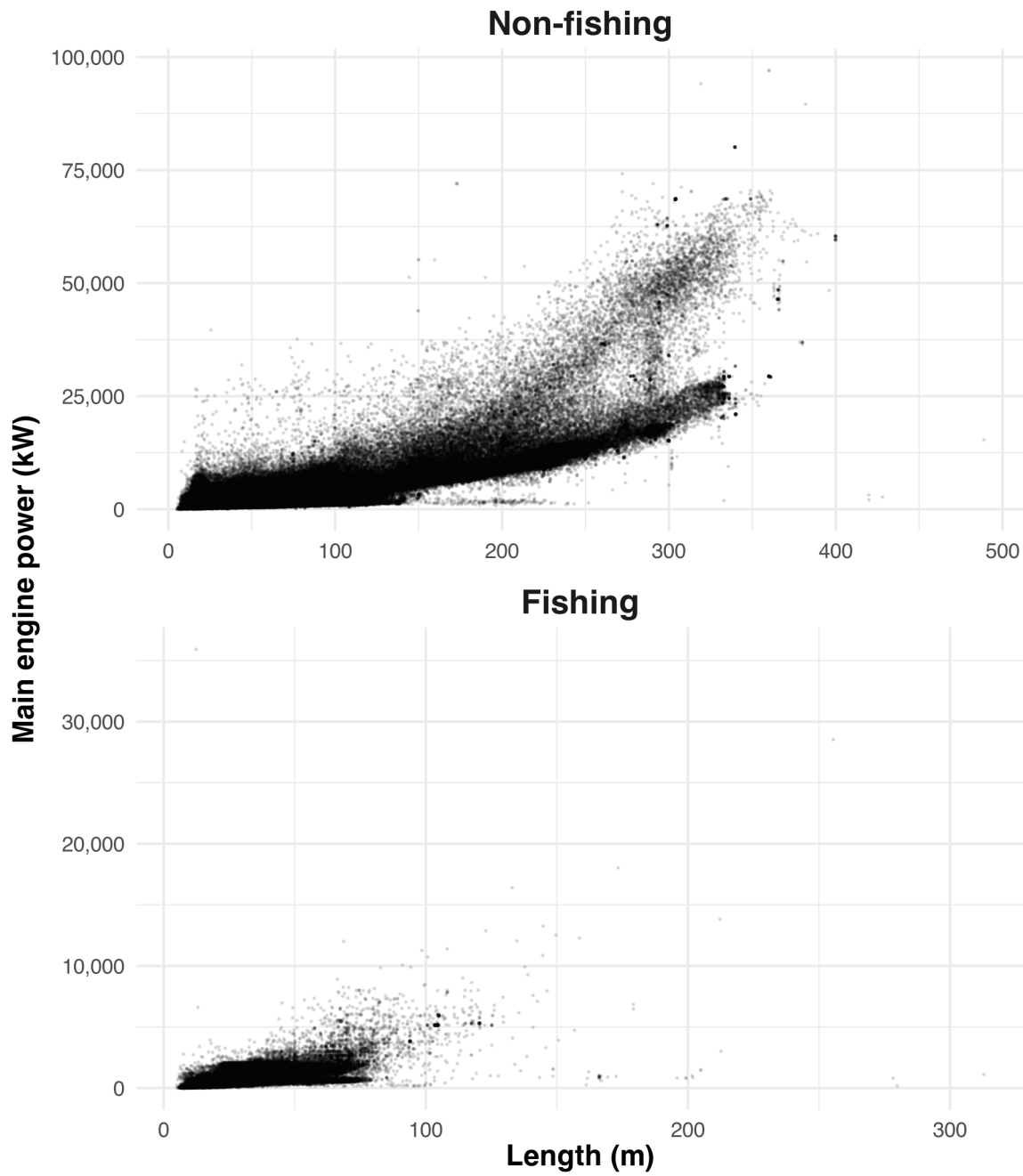


Figure 11: Relationship between main engine power (kilowatts) and length (meters) for all AIS-broadcasting vessels, disaggregated by non-fishing and fishing vessels.

4.11 Table 3

Table 5: Data sources for all included model features, including which random forest model uses the features (M1: detections classification; M2: detections regression; and/or M3: emissions regression), the units, feature type, whether it varies spatially, whether it varies temporally, and the source name and reference citation.

Model feature	Category	M1	M2	M3	Units	Type	Spatial	Temporal	Source	Ref.
Presence/absence of unmatched S1 detections	Dependent variable	Yes	No	No	-	Boolean	Yes	Yes	GFW	[?]
Unmatched S1 detections per km ² imaged	Dependent variable	No	Yes	No	detects/km ²	Numeric	Yes	Yes	GFW	[?]
Log(AIS emissions)	Dependent variable	No	No	Yes	log(MT)	Numeric	Yes	Yes	GFW	-
Vessel type (fishing or non-fishing)	Dependent variable	Yes	Yes	Yes	-	Boolean	Yes	Yes	GFW	[?]
Lower threshold of vessel length bin	S1 detection features	Yes	Yes	No	m	Numeric	No	No	GFW	[?]
Vessel length bin category	S1 detection features	Yes	Yes	No	-	Factor	Yes	Yes	GFW	[?]
S1 detections per km ² of area for each length bin	S1 detection features	No	No	Yes	detects/km ²	Numeric	Yes	Yes	GFW	[?]
Total pixel area imaged by S1 across all time	S1 detection features	Yes	Yes	No	km ²	Numeric	Yes	No	GFW	[?]
Unique AIS-broadcasting vessels in pixel-month	AIS reception quality	Yes	Yes	No	-	Numeric	Yes	Yes	GFW	[?]
Unique AIS-broadcasting vessels globally	AIS reception quality	Yes	Yes	No	-	Numeric	No	Yes	GFW	[?]
Positions reported by AIS type A	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	No	GFW	[?]
Positions reported by AIS type B	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	No	GFW	[?]
Proportion of messages, dynamic AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFW	[?]
Proportion of messages, terrestrial AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFW	[?]
Proportion of messages, satellite AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFW	[?]
Month of year	Temporal features	Yes	Yes	Yes	-	Factor	No	Yes	GFW	-
Post-2020 IMO Sulphur policy	Temporal features	No	No	Yes	-	Boolean	No	Yes	IMO	[?]
Ocean basin	Geographic features	Yes	Yes	Yes	-	Factor	Yes	No	GFW	[?]
Pixel water area	Geographic features	Yes	Yes	Yes	km ²	Numeric	Yes	No	GFW	[?]
Pixel water fraction	Geographic features	No	No	Yes	-	Numeric	Yes	No	GFW	[?]
Mean ocean depth	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFW	[?]
Mean distance from shore	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFW	[?]
Mean distance from nearest port	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFW	[?]
Minimum distance to top 20 port	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	Lloyd's List	[?]
Mean wind speed	Weather features	No	Yes	No	m/s	Numeric	Yes	Yes	ERA5	[?]
Mean significant wave and swell height	Weather features	No	Yes	No	m	Numeric	Yes	Yes	ERA5	[?]
Mean wave period	Weather features	No	Yes	No	s	Numeric	Yes	Yes	ERA5	[?]
Mean sea surface temperature	Weather features	No	Yes	No	K	Numeric	Yes	Yes	ERA5	[?]
Mean sea level pressure	Weather features	No	Yes	No	Pa	Numeric	Yes	Yes	ERA5	[?]
Total precipitation	Weather features	No	Yes	No	m	Numeric	Yes	Yes	ERA5	[?]

4.12 Figure 12

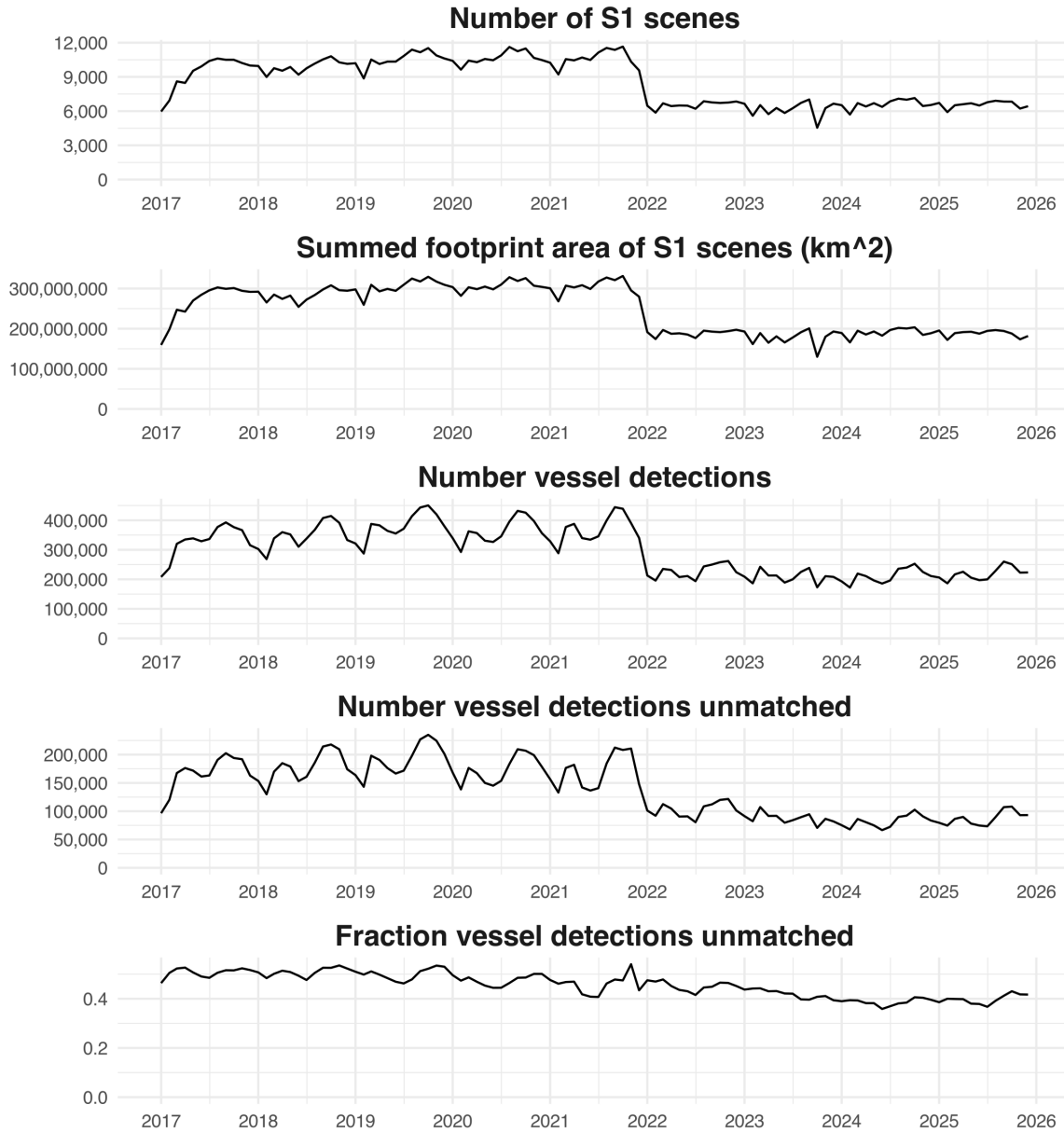


Figure 12: Monthly statistics of global S1 coverage including total number of S1 scenes, total imaged area summed across all S1 scenes (which counts each revisit to a given area separately), total number of vessel detections, total number of vessel detections that could not be matched to an AIS-broadcasting vessels, and fraction of the total number of vessel detections that could not be matched to an AIS-broadcasting vessels.

4.13 Figure 13

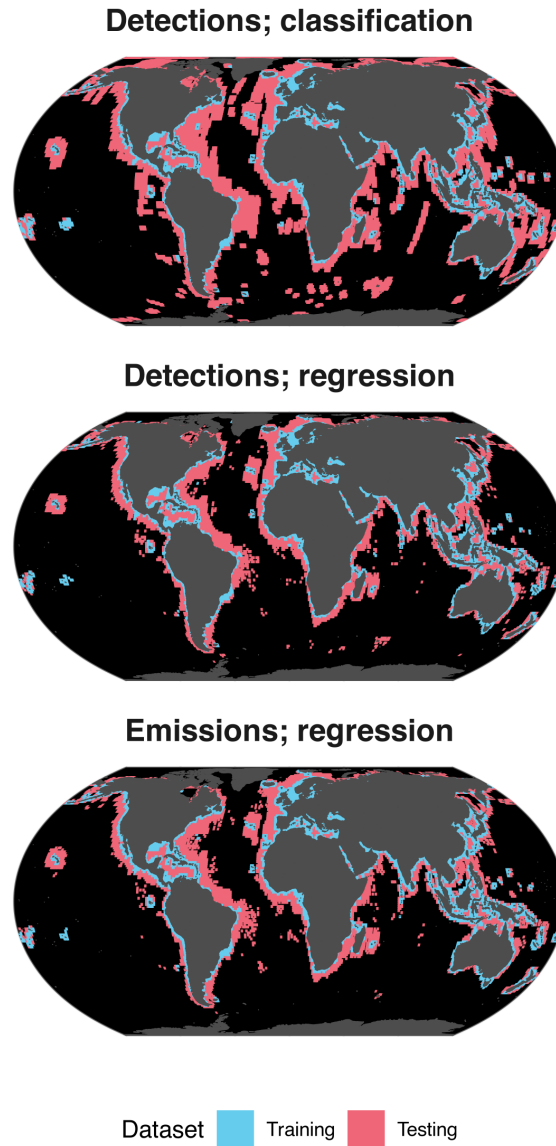


Figure 13: Maps of pixels that were used in the training and testing datasets for the simulated outside S1 footprint performance tests for each model. Pixels $\leq 100\text{km}$ from the shore were used for training; pixels $> 100\text{km}$ from shore were used for testing.

4.14 Figure 14

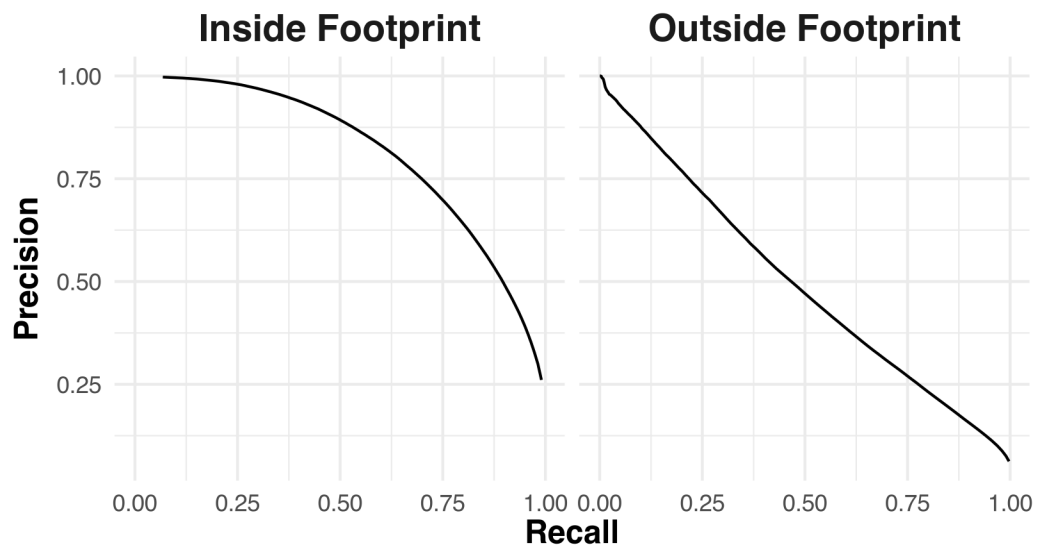


Figure 14: Precision-recall curves of the detection presence/absence classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

4.15 Figure 15

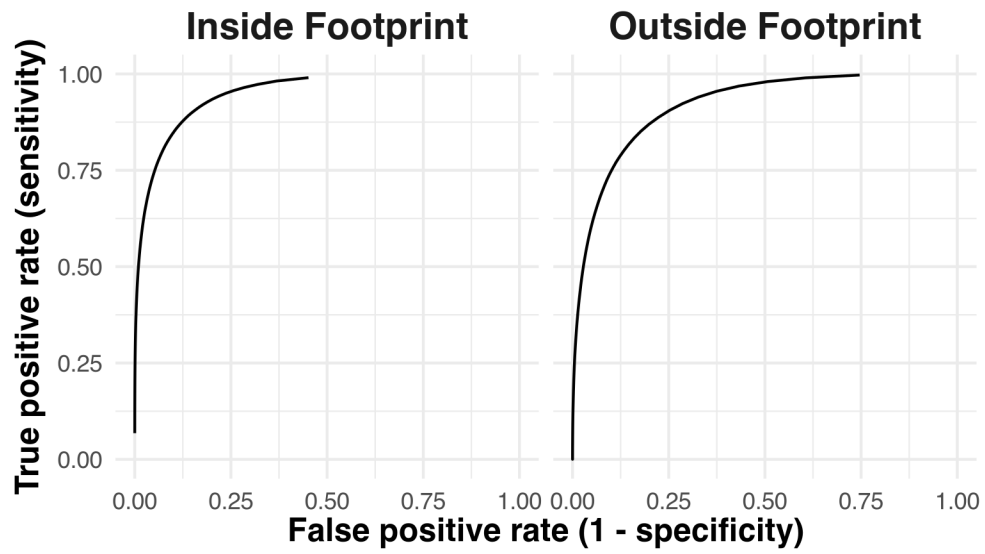


Figure 15: Receiver Operating Characteristic (ROC) curves of the detection classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

4.16 Figure 16

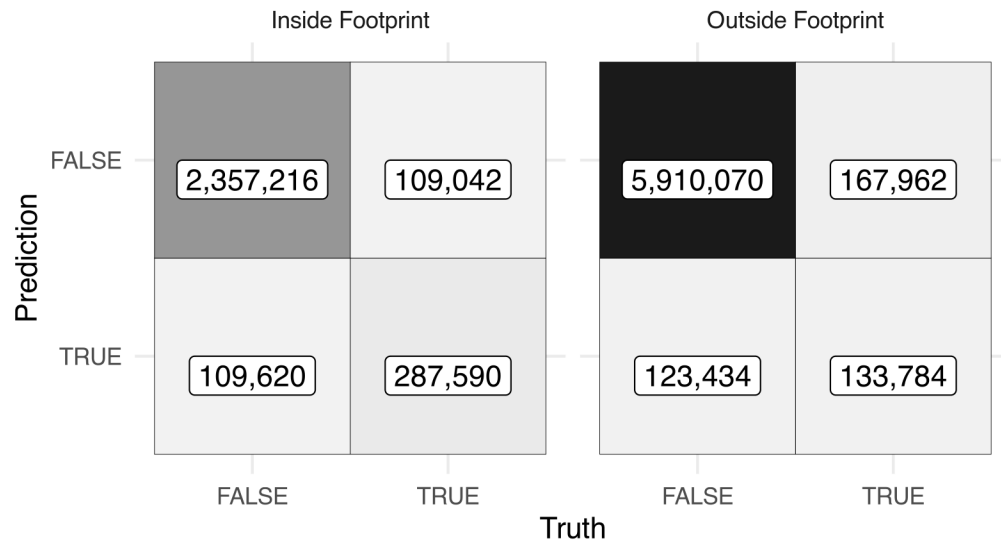


Figure 16: Confusion matrices of the detection classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

Table 6: Summary of dark fleet model performance metrics for the: 1) emissions regression model; 2) detections classification mode; and 3) detections regression model. Performance estimates for each metric are shown for both the inside S1 footprint and outside S1 footprint tests. All metrics are calculated at the unit of observation except for 'Total percent error', which is the percentage error in the total sum of all predictions relative to the total sum of all truth values.

Model	Pollutant	Metric	Inside footprint	Outside footprint
Emissions regression	CO2	Traditional R ²	0.94	0.73
Emissions regression	CH4	Traditional R ²	0.90	0.63
Emissions regression	N2O	Traditional R ²	0.94	0.71
Emissions regression	CO	Traditional R ²	0.92	0.63
Emissions regression	NOX	Traditional R ²	0.87	0.57
Emissions regression	SOX	Traditional R ²	0.95	0.73
Emissions regression	PM2.5	Traditional R ²	0.94	0.65
Emissions regression	PM10	Traditional R ²	0.94	0.65
Emissions regression	VOCs	Traditional R ²	0.88	0.60
Emissions regression	CO2	Total percent error	-8.61	-10.08
Emissions regression	CH4	Total percent error	-20.26	-30.01
Emissions regression	N2O	Total percent error	-10.70	-15.51
Emissions regression	CO	Total percent error	-19.10	-29.29
Emissions regression	NOX	Total percent error	-23.68	-36.44
Emissions regression	SOX	Total percent error	-8.23	-9.01
Emissions regression	PM2.5	Total percent error	-16.21	-25.52
Emissions regression	PM10	Total percent error	-16.21	-25.52
Emissions regression	VOCs	Total percent error	-22.23	-32.45
Detections classification	-	Accuracy	0.92	0.95
Detections classification	-	F1 score	0.72	0.48
Detections classification	-	Precision	0.72	0.52
Detections classification	-	Precision-recall AUC	0.81	0.49
Detections classification	-	Recall	0.73	0.44
Detections classification	-	ROC AUC	0.95	0.92
Detections classification	-	Sensitivity	0.73	0.44
Detections classification	-	Specificity	0.96	0.98
Detections regression	-	Traditional R ²	0.73	0.46
Detections regression	-	Total percent error	1.18	28.46

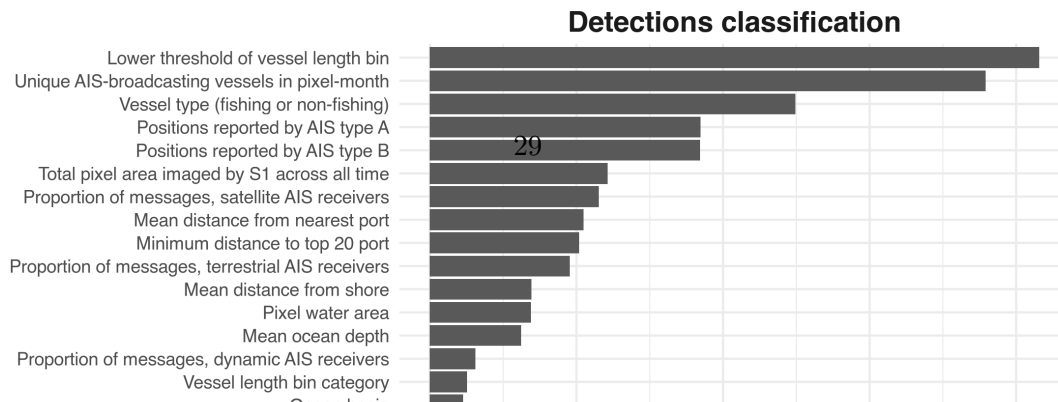
Table 7: Linear regression fit statistics for the final models trained to estimate emissions of non-CO₂ gases from CO₂ emissions. The model for each gas is a zero-intercept linear regression with 5 terms: CO₂ emissions interacted with each combination of period (pre/post-2020 IMO sulphur policy) and vessel type (fishing/non-fishing), plus CO₂ emissions interacted with the log of mean distance from shore.

Pollutant	Term	Estimate	SE	T-stat.	p-value
CH4	CO2 : log(distance from shore + 1)	1.25e-06	3.29e-09	379.9	<0.001
CH4	CO2 : pre-2020 : non-fishing	-1.21e-06	3.17e-08	-38.0	<0.001
CH4	CO2 : post-2020 : non-fishing	-1.92e-06	3.14e-08	-61.3	<0.001
CH4	CO2 : pre-2020 : fishing	3.69e-06	8.42e-08	43.9	<0.001
CH4	CO2 : post-2020 : fishing	3.73e-06	4.52e-08	82.6	<0.001
N2O	CO2 : log(distance from shore + 1)	7.13e-07	2.32e-09	308.0	<0.001
N2O	CO2 : pre-2020 : non-fishing	4.00e-05	2.23e-08	1869.4	<0.001
N2O	CO2 : post-2020 : non-fishing	4.00e-05	2.21e-08	1865.7	<0.001
N2O	CO2 : pre-2020 : fishing	4.00e-05	5.92e-08	727.0	<0.001
N2O	CO2 : post-2020 : fishing	4.00e-05	3.18e-08	1379.9	<0.001
CO	CO2 : log(distance from shore + 1)	6.00e-05	1.57e-07	382.3	<0.001
CO	CO2 : pre-2020 : non-fishing	3.00e-05	1.51e-06	19.2	<0.001
CO	CO2 : post-2020 : non-fishing	-5.49e-06	1.50e-06	-3.7	<0.001
CO	CO2 : pre-2020 : fishing	2.70e-04	4.02e-06	67.9	<0.001
CO	CO2 : post-2020 : fishing	2.80e-04	2.16e-06	128.1	<0.001
NOX	CO2 : log(distance from shore + 1)	2.02e-03	4.33e-06	466.7	<0.001
NOX	CO2 : pre-2020 : non-fishing	-7.34e-03	4.00e-05	-176.3	<0.001
NOX	CO2 : post-2020 : non-fishing	-8.45e-03	4.00e-05	-205.2	<0.001
NOX	CO2 : pre-2020 : fishing	-9.70e-04	1.10e-04	-8.8	<0.001
NOX	CO2 : post-2020 : fishing	-2.14e-03	6.00e-05	-36.1	<0.001
SOX	CO2 : log(distance from shore + 1)	-1.00e-05	7.15e-08	-173.1	<0.001
SOX	CO2 : pre-2020 : non-fishing	6.30e-03	6.88e-07	9164.2	<0.001
SOX	CO2 : post-2020 : non-fishing	1.66e-03	6.81e-07	2441.1	<0.001
SOX	CO2 : pre-2020 : fishing	6.37e-03	1.83e-06	3486.1	<0.001
SOX	CO2 : post-2020 : fishing	1.67e-03	9.81e-07	1705.6	<0.001
PM2.5	CO2 : log(distance from shore + 1)	4.00e-05	9.11e-08	401.0	<0.001
PM2.5	CO2 : pre-2020 : non-fishing	4.10e-04	8.76e-07	472.5	<0.001
PM2.5	CO2 : post-2020 : non-fishing	5.00e-05	8.67e-07	58.1	<0.001
PM2.5	CO2 : pre-2020 : fishing	5.30e-04	2.33e-06	228.9	<0.001
PM2.5	CO2 : post-2020 : fishing	2.10e-04	1.25e-06	168.4	<0.001
PM10	CO2 : log(distance from shore + 1)	4.00e-05	9.90e-08	401.0	<0.001
PM10	CO2 : pre-2020 : non-fishing	4.50e-04	9.52e-07	472.5	<0.001
PM10	CO2 : post-2020 : non-fishing	5.00e-05	9.43e-07	58.1	<0.001
PM10	CO2 : pre-2020 : fishing	5.80e-04	2.53e-06	228.9	<0.001
PM10	CO2 : post-2020 : fishing	2.30e-04	1.36e-06	168.4	<0.001
VOCs	CO2 : log(distance from shore + 1)	1.00e-04	2.02e-07	495.7	<0.001
VOCs	CO2 : pre-2020 : non-fishing	-3.90e-04	1.94e-06	-203.0	<0.001
VOCs	CO2 : post-2020 : non-fishing	-4.40e-04	1.92e-06	-228.0	<0.001
VOCs	CO2 : pre-2020 : fishing	-2.00e-04	5.15e-06	-39.0	<0.001
VOCs	CO2 : post-2020 : fishing	-2.10e-04	2.77e-06	-75.7	<0.001

4.17 Table 4

4.18 Table 5

4.19 Figure 17



4.20 Figure 18

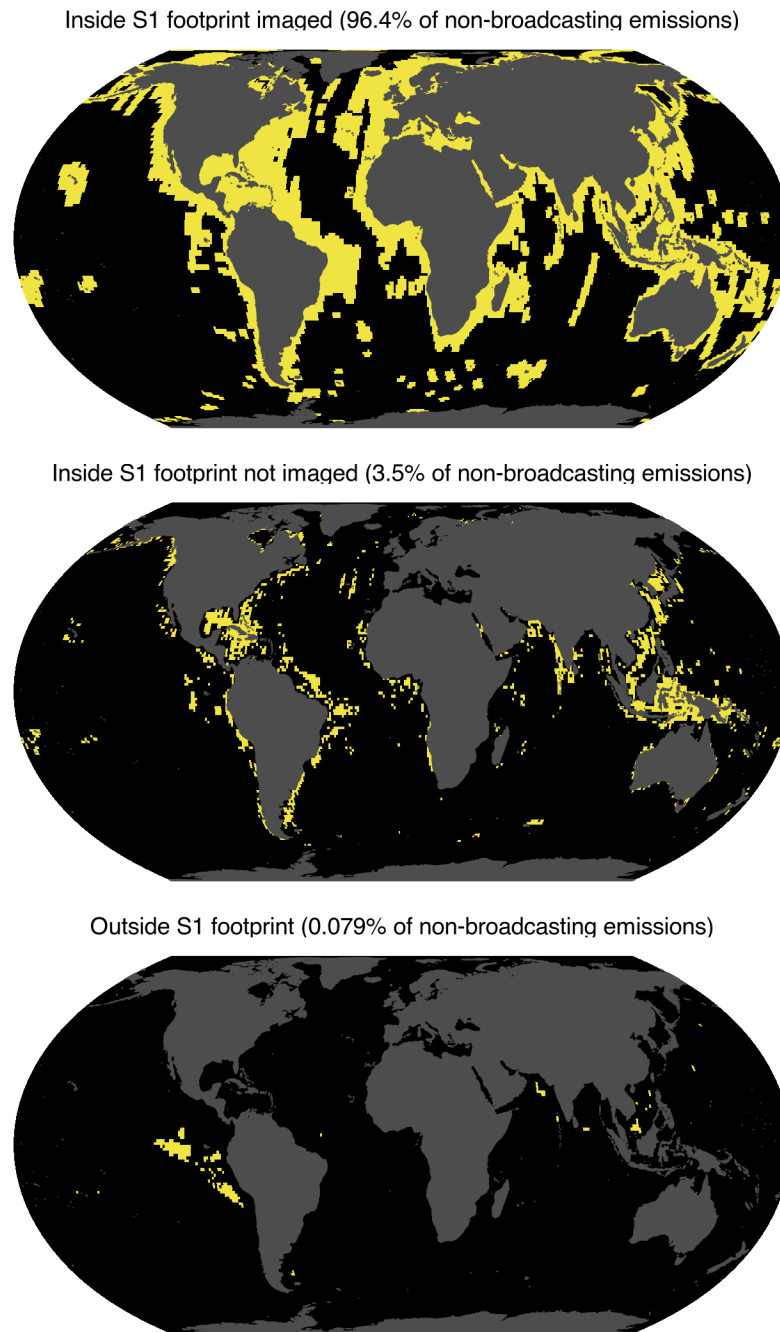


Figure 18: Spatial coverage of the non-broadcasting emissions model. Areas are categorized according to whether they fall inside or outside the S1 footprint and whether they have been imaged by S1 at any point in time. A given pixel may be imaged in some months but not in others. Pixels outside the footprint have never been imaged. The percentage of total CO₂ emissions from non-broadcasting vessels attributed to each category is shown.

5 Supplementary materials: Supplementary figures

5.1 Figure S1

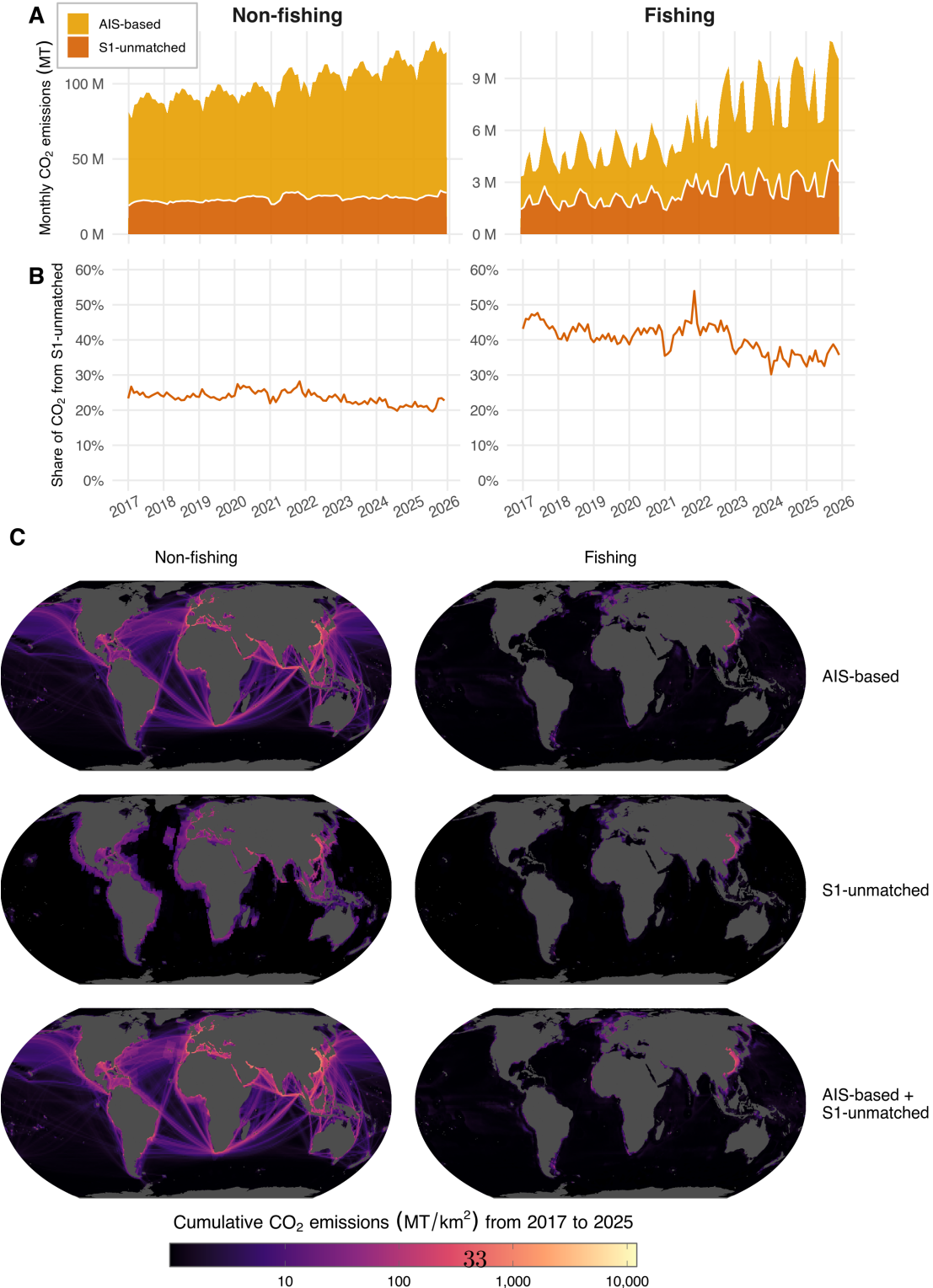


Figure 19: (A) Time series of monthly global CO₂ emissions, shown for both AIS-broadcasting and S1-detected non-broadcasting vessels, and for both fishing and non-fishing vessels. (B) Time series of the monthly share of global CO₂ emissions attributable to S1-detected non-broadcasting vessels, for both fishing and non-fishing vessels. (C) Spatial distribution of cumulative 2017-2025 CO₂ emissions from AIS-broadcasting

5.2 Figure S2

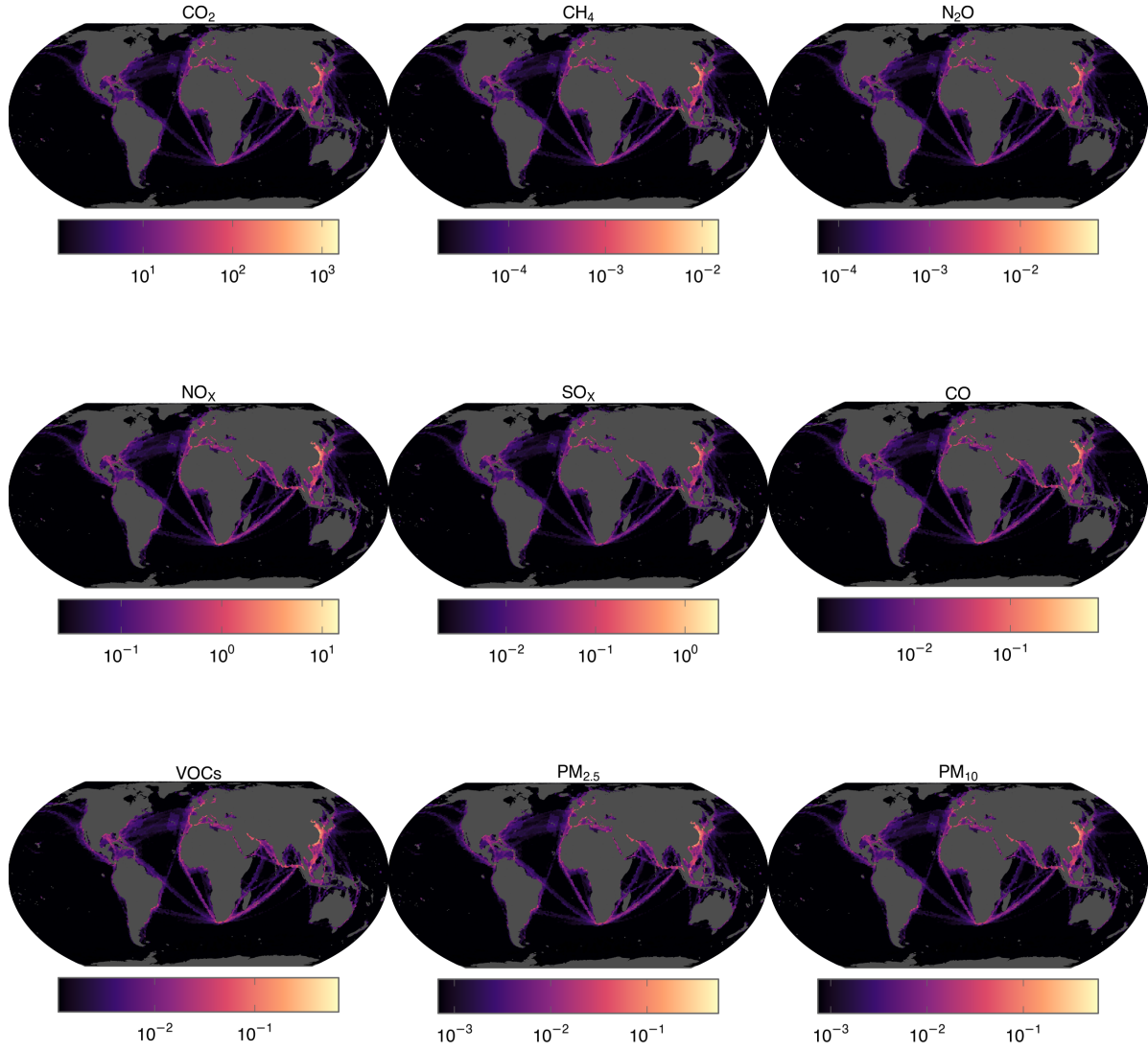


Figure 20: Spatial maps of 2025 emissions for each GHG and pollutant, aggregated across AIS-broadcasting vessels and non-broadcasting vessels, and shown at a 1x1 degree spatial resolution. Emissions are area-normalized for each pixel (metric tonnes per square kilometer) and shown using a log-10 scale. Each panel is scaled independently, spanning the 75th percentile to the maximum of that pollutant, with lower values drawn in the darkest colour; colours are therefore not comparable between panels.

5.3 Figure S3

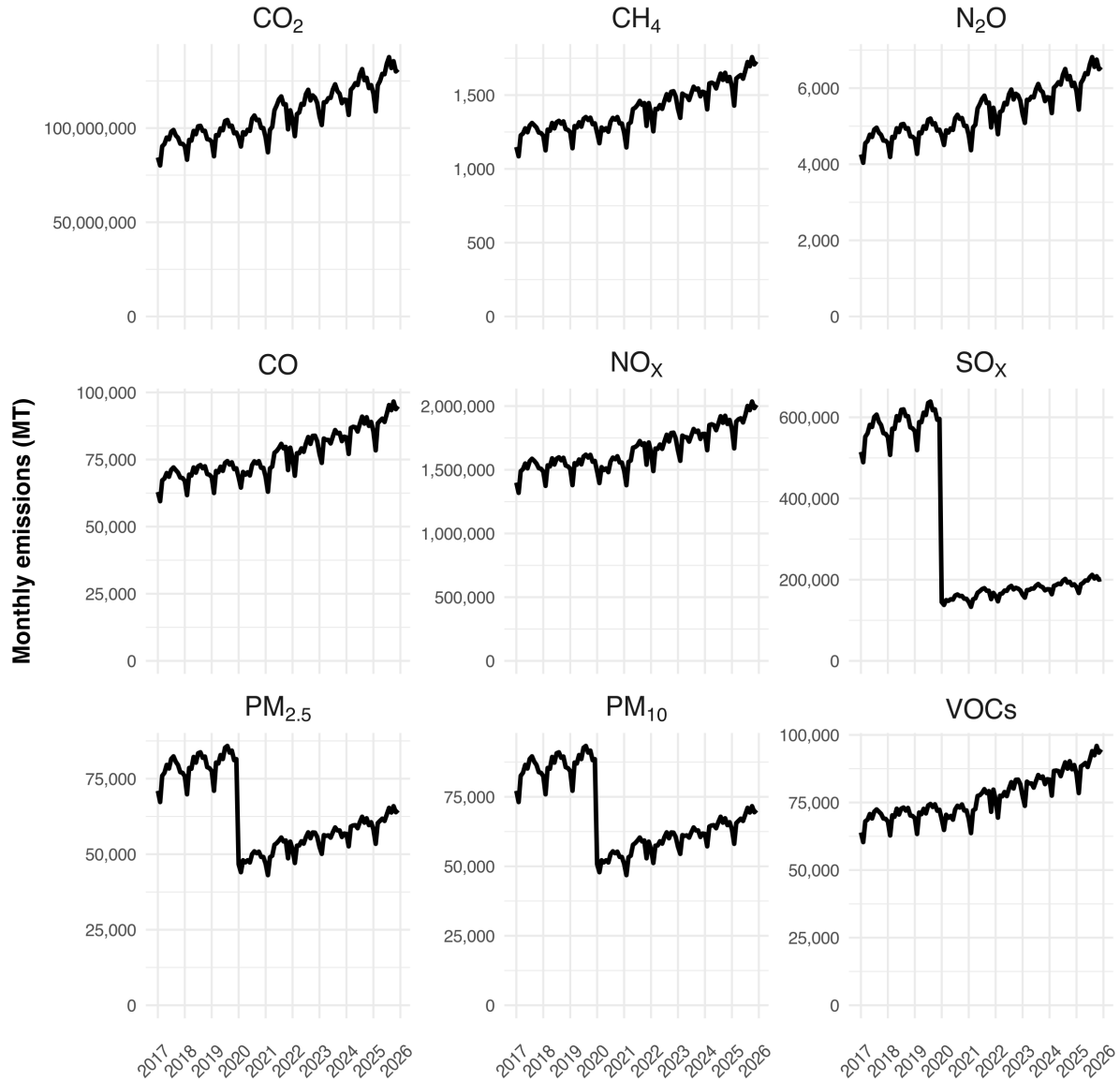


Figure 21: Time series of monthly global emissions for each GHG and pollutant, aggregated across AIS-broadcasting vessels and non-broadcasting vessels for which we only have Sentinel-1 data.

5.4 Figure S4

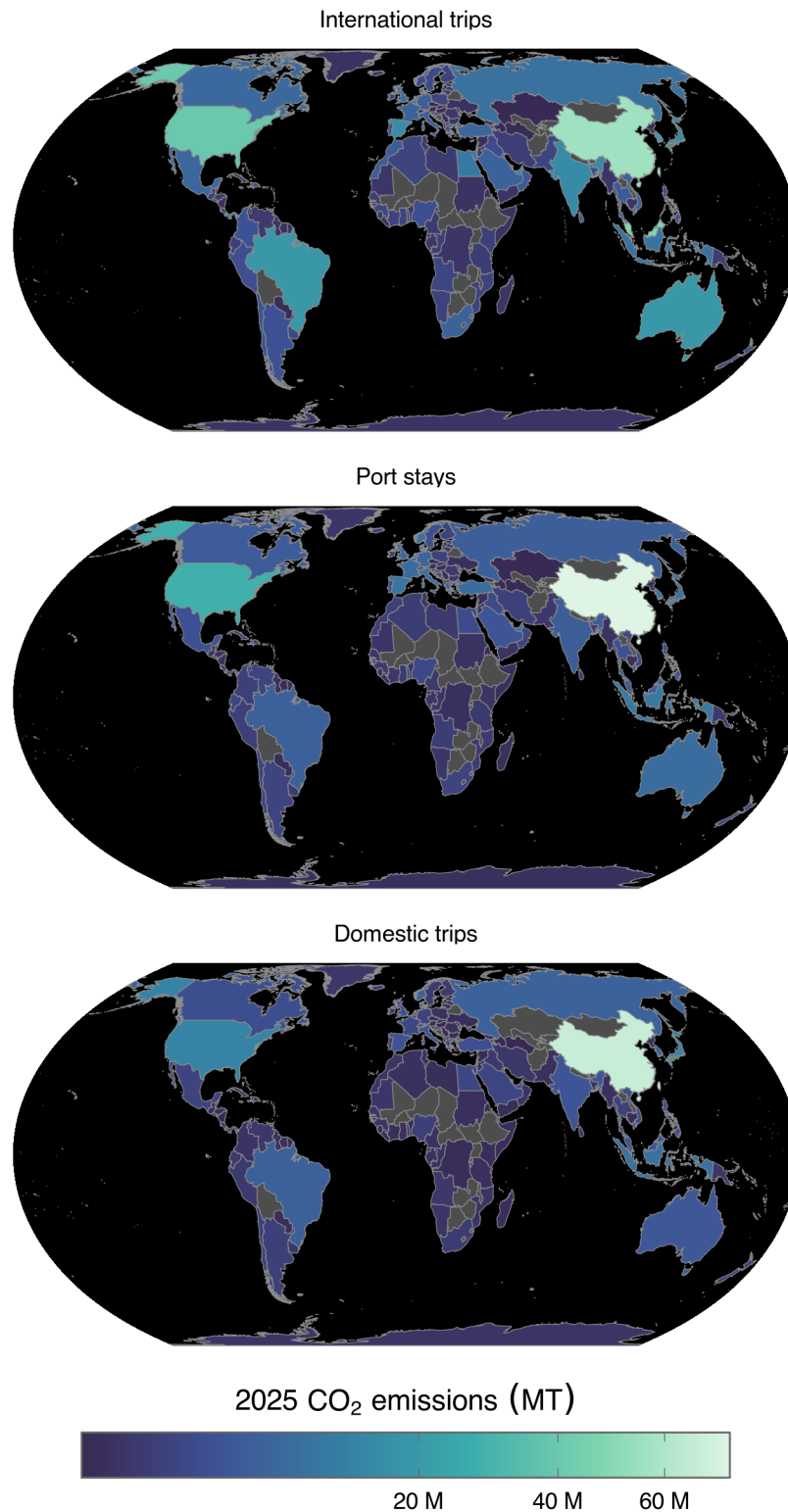


Figure 22: CO₂ emissions by country for AIS7-broadcasting vessels during different activity types (international trips, port stays, and domestic trips), aggregated for activities ending in 2025. Emissions for each international trip are partitioned equally to its departure and arrival country. The color scale is square-root transformed, so that lower-emitting countries remain distinguishable alongside the largest emitters. Countries with no values are shown in gray.

5.5 Figure S5

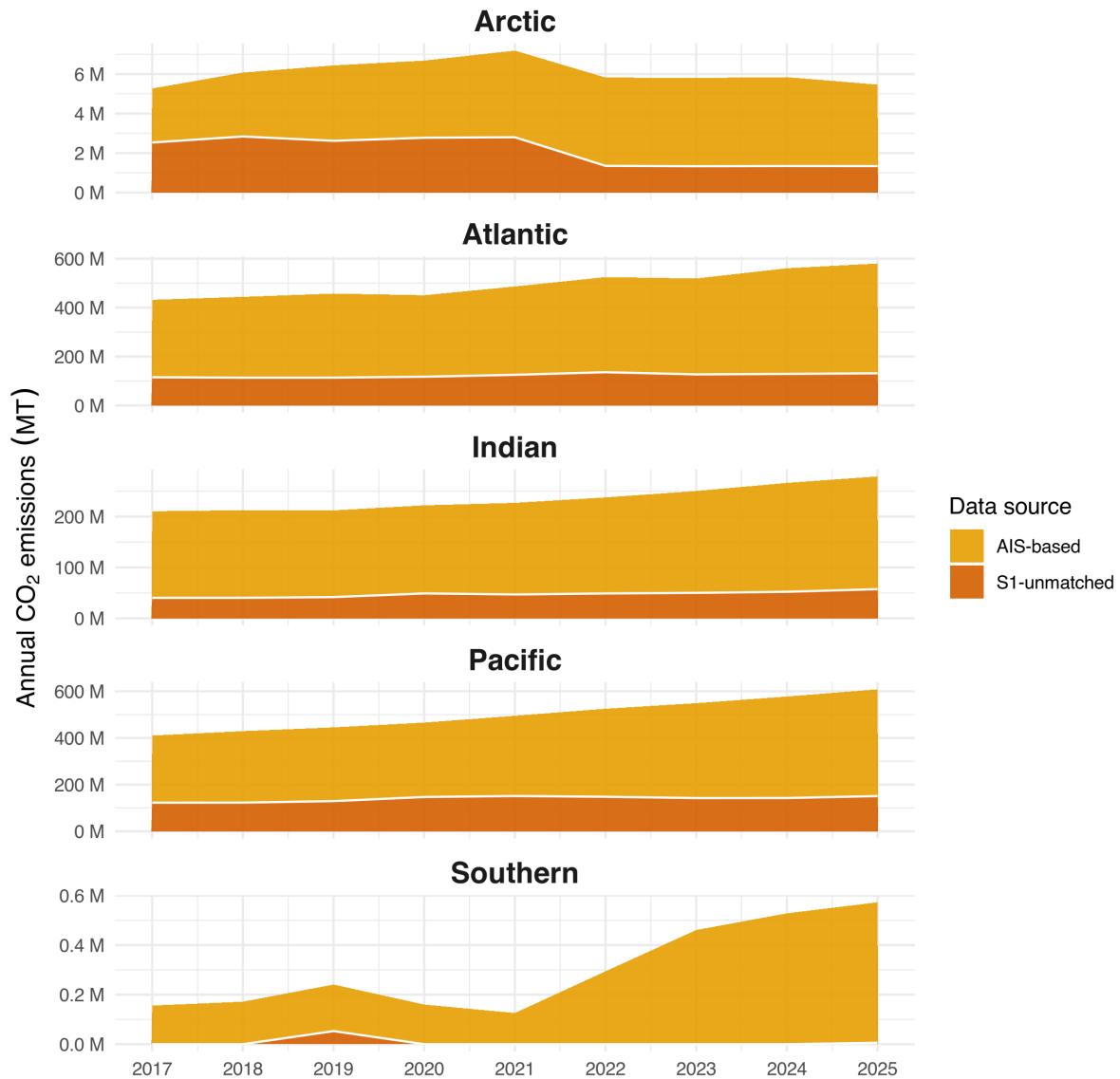


Figure 23: Annual total CO₂ emissions by ocean, disaggregated by data source (AIS-broadcasting vessels, and non-broadcasting vessels detected by S1).

6 Supplementary materials: Supplementary tables

6.1 Table S1

Table 8: Summary of 2017 and 2025 emissions, data source and pollutant.

Source	Pollutant	2017 emissions, MT	2025 emissions, MT	Percent 2025 total	2017 to 2025 change
AIS-based	CO2	8.24e+08	1.19e+09	78%	44%
S1-unmatched	CO2	2.83e+08	3.43e+08	22%	21%
AIS-based + S1-unmatched	CO2	1.11e+09	1.53e+09	100%	38%
AIS-based	CH4	11,500	15,800	80%	37%
S1-unmatched	CH4	3,420	3,990	20%	17%
AIS-based + S1-unmatched	CH4	14,900	19,700	100%	33%
AIS-based	N2O	41,800	59,500	78%	42%
S1-unmatched	N2O	13,900	16,800	22%	21%
AIS-based + S1-unmatched	N2O	55,700	76,300	100%	37%
AIS-based	CO	628,000	865,000	80%	38%
S1-unmatched	CO	189,000	222,000	20%	17%
AIS-based + S1-unmatched	CO	817,000	1,090,000	100%	33%
AIS-based	NOX	14,100,000	18,500,000	80%	31%
S1-unmatched	NOX	3,960,000	4,490,000	20%	14%
AIS-based + S1-unmatched	NOX	18,100,000	2.3e+07	100%	27%
AIS-based	SOX	5,030,000	1,830,000	78%	-64%
S1-unmatched	SOX	1,750,000	527,000	22%	-70%
AIS-based + S1-unmatched	SOX	6,770,000	2,360,000	100%	-65%
AIS-based	PM2.5	701,000	590,000	79%	-16%
S1-unmatched	PM2.5	226,000	153,000	21%	-32%
AIS-based + S1-unmatched	PM2.5	927,000	743,000	100%	-20%
AIS-based	PM10	762,000	641,000	79%	-16%
S1-unmatched	PM10	246,000	166,000	21%	-32%
AIS-based + S1-unmatched	PM10	1,010,000	807,000	100%	-20%
AIS-based	VOCs	641,000	868,000	80%	35%
S1-unmatched	VOCs	185,000	213,000	20%	15%
AIS-based + S1-unmatched	VOCs	825,000	1,080,000	100%	31%

6.2 Table S2

Table 9: Underestimation of 2025 emissions using only AIS data, and overestimation of 2017 to 2025 emissions trend using only AIS data, by pollutant.

Pollutant	Underestimation of 2025 emissions	Overestimation of 2017-2025 emissions trend
CO2	22%	15%
CH4	20%	15%
N2O	22%	15%
CO	20%	14%
NOX	20%	14%
SOX	22%	-2%
PM2.5	21%	-20%
PM10	21%	-20%
VOCs	20%	15%

Table 10: For each ocean, summary of: 2025 total (AIS+S1) CO₂ emissions (metric tonnes); percentage of global total (AIS+S1) 2025 CO₂ emissions from across all oceans; percent of 2025 total (AIS+S1) CO₂ emissions by non-broadcasting vessels; percent change in total (AIS+S1) CO₂ emissions from 2017 to 2025; and the percentage of all observations in the dataset (pixel-months) from 2017-2025 that were imaged by S1.

Summary statistic	Atlantic	Southern	Pacific	Indian	Arctic
2025 total CO2 emissions (million MT)	449	0.567	457	222	4.13
Percent of total CO2 2025 emissions	39.37%	0.04%	41.28%	18.93%	0.37%
Percent of total CO2 emissions by non-broadcasting	22.64%	1.02%	24.81%	20.58%	24.50%
Percent change in total CO2 emissions (2017 to 2025)	34.31%	267.19%	48.27%	32.49%	3.74%
Percent observations imaged by S1 (across 2017-2025)	35.94%	0.06%	17.74%	16.99%	13.55%

6.3 Table S3

6.4 Table S4

Table 11: For each data source, summary of: total 2024 CO₂ emissions (metric tonnes); and percent change in CO₂ emissions from 2017 to 2024.

Data source	2024 emissions, billion MT	Change in emissions, 2017 to 2024
Other transportation sectors (EDGAR)	7.44	3.35%
All sectors (EDGAR)	39.6	7.06%
Marine vessels (AIS-based + S1-unmatched)	1.47	32.49%

6.5 Table S5

Table 12: For both non-fishing vessels and fishing vessels, we summarize: the total emissions (metric tonnes) of each GHG gas; the social cost associated with each GHG (USD/metric tonne); the social cost of each (USD); the total social cost summed across all three gases (USD). We use 2021 for non-fishing vessels since it is the year for which we have the total value of all traded goods at sea. We use 2022 for fishing vessels since it is the year for which we have the total value of wild capture fisheries.

Year	Vessel type	GHG	Emissions (MT)	SC (USD/MT)	SC (billion USD)
2021	Non-fishing	CH4	15,296	2,900	0.044
2021	Non-fishing	CO2	1,205,568,569	283	341.176
2021	Non-fishing	N2O	60,152	49,600	2.984
Total	Non-fishing	-	-	-	344.204
2022	Fishing	CH4	1,373	2,900	0.004
2022	Fishing	CO2	82,662,425	283	23.393
2022	Fishing	N2O	4,211	49,600	0.209
Total	Fishing	-	-	-	23.606

7 Supplementary materials: Supplementary figures: Comparing our emission estimates to other inventories

7.1 Figure S6

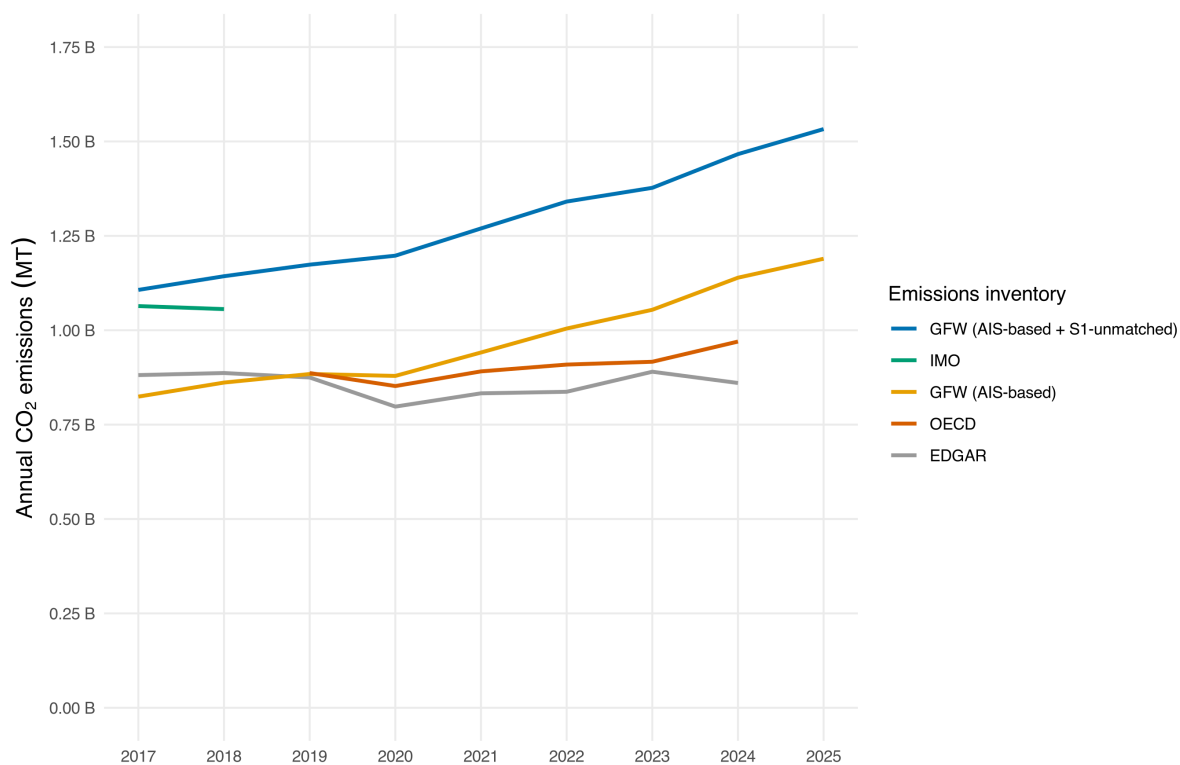


Figure 24: Comparison of annual marine vessel CO₂ emissions inventory estimates.

Table 13: Top 30 countries by 2025 CO₂ emissions (millions of metric tonnes) from AIS-broadcasting vessels, shown separately for each activity type, with all remaining countries pooled into a single Other row. Emissions for each international trip are partitioned equally to its departure and arrival country. These are the totals mapped in Figure ??.

International trips		Port stays		Domestic trips	
Country	CO ₂	Country	CO ₂	Country	CO ₂
China	55.5	China	74.0	China	66.8
Malaysia	48.1	United States	27.8	Japan	13.1
United States	39.9	Indonesia	10.6	United States	13.1
South Korea	20.2	Spain	9.8	Indonesia	8.8
Australia	19.1	South Korea	9.0	South Korea	5.7
Brazil	18.9	Netherlands	9.0	Russia	5.6
Spain	15.7	Japan	8.9	Norway	5.4
Japan	15.5	United Arab Emirates	8.4	Brazil	5.2
India	15.2	Norway	8.4	Australia	3.8
United Arab Emirates	14.2	Italy	8.1	Malaysia	3.4
Panama	13.8	Australia	7.6	India	3.4
Singapore	12.7	Turkey	7.4	Italy	3.2
Egypt	10.4	France	7.2	Spain	3.1
Indonesia	9.0	United Kingdom	6.7	Turkey	2.7
Russia	8.5	Germany	6.0	United Kingdom	2.6
United Kingdom	7.7	Greece	5.8	Canada	2.5
Netherlands	7.4	Brazil	5.3	Vietnam	2.3
Canada	6.9	Russia	5.0	United Arab Emirates	2.2
Mexico	6.4	India	4.8	Philippines	2.2
Turkey	6.3	Malaysia	4.7	Greece	1.8
Taiwan	6.1	Canada	4.5	Taiwan	1.8
Belgium	6.0	Denmark	4.4	France	1.7
France	5.6	Belgium	3.9	Egypt	1.7
Saudi Arabia	5.5	Saudi Arabia	3.3	Saudi Arabia	1.6
South Africa	5.5	Mexico	2.8	Mexico	1.3
Vietnam	5.4	Thailand	2.8	Germany	1.1
Italy	5.0	Sweden	2.5	Chile	1.1
Germany	5.0	Philippines	2.3	Thailand	1.1
Sri Lanka	4.9	Taiwan	2.2	New Zealand	1.0
Morocco	4.7	Egypt	2.2	Argentina	1.0
Other	107.2	Other	59.3	Other	19.5

Table 14: Comparison of annual marine vessel CO₂ emissions inventory estimates (billions of metric tonnes).

Year	GFW (AIS-based + S1-unmatched)	GFW (AIS-based)	EDGAR	OECD	IMO
2017	1.11	0.824	0.881	-	1.06
2018	1.14	0.861	0.887	-	1.06
2019	1.17	0.884	0.875	0.887	-
2020	1.2	0.879	0.798	0.852	-
2021	1.27	0.941	0.833	0.891	-
2022	1.34	1	0.837	0.909	-
2023	1.38	1.05	0.89	0.917	-
2024	1.47	1.14	0.86	0.97	-
2025	1.53	1.19	-	-	-

7.2 Table S6

7.3 Table S7

7.4 Figure S7

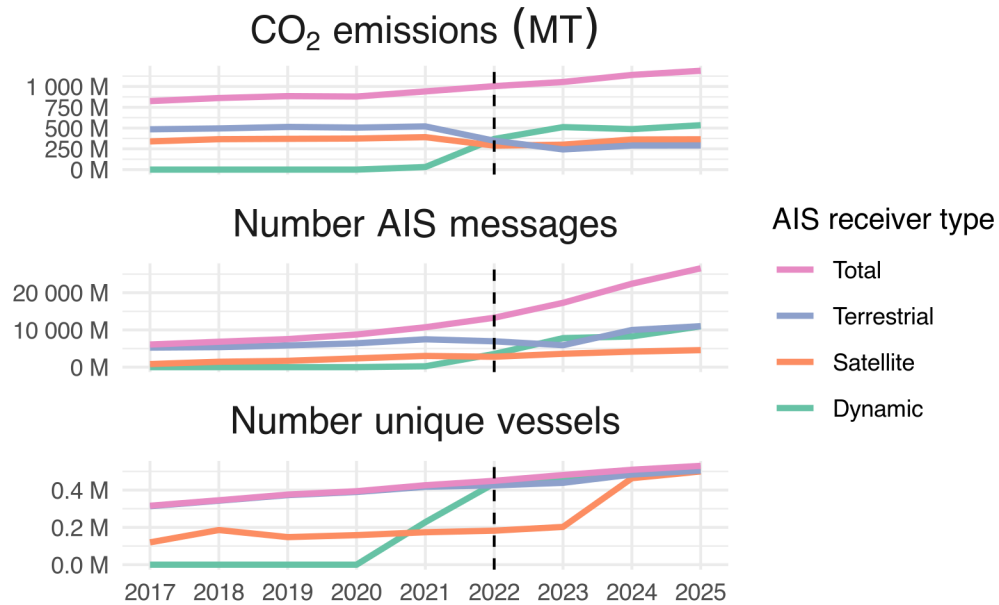


Figure 25: Annual total CO₂ emissions (metric tonnes) by AIS-broadcasting vessels, AIS pings (i.e., individual vessel timestamped location messages), and number of unique AIS-broadcasting vessels. Trends are disaggregated by AIS receiver type (satellite, terrestrial, and dynamic), and also shown as a total aggregated across receiver types. A dashed vertical line is shown for 2022, the first year in which dynamic AIS was widely adopted.

7.5 Figure S8

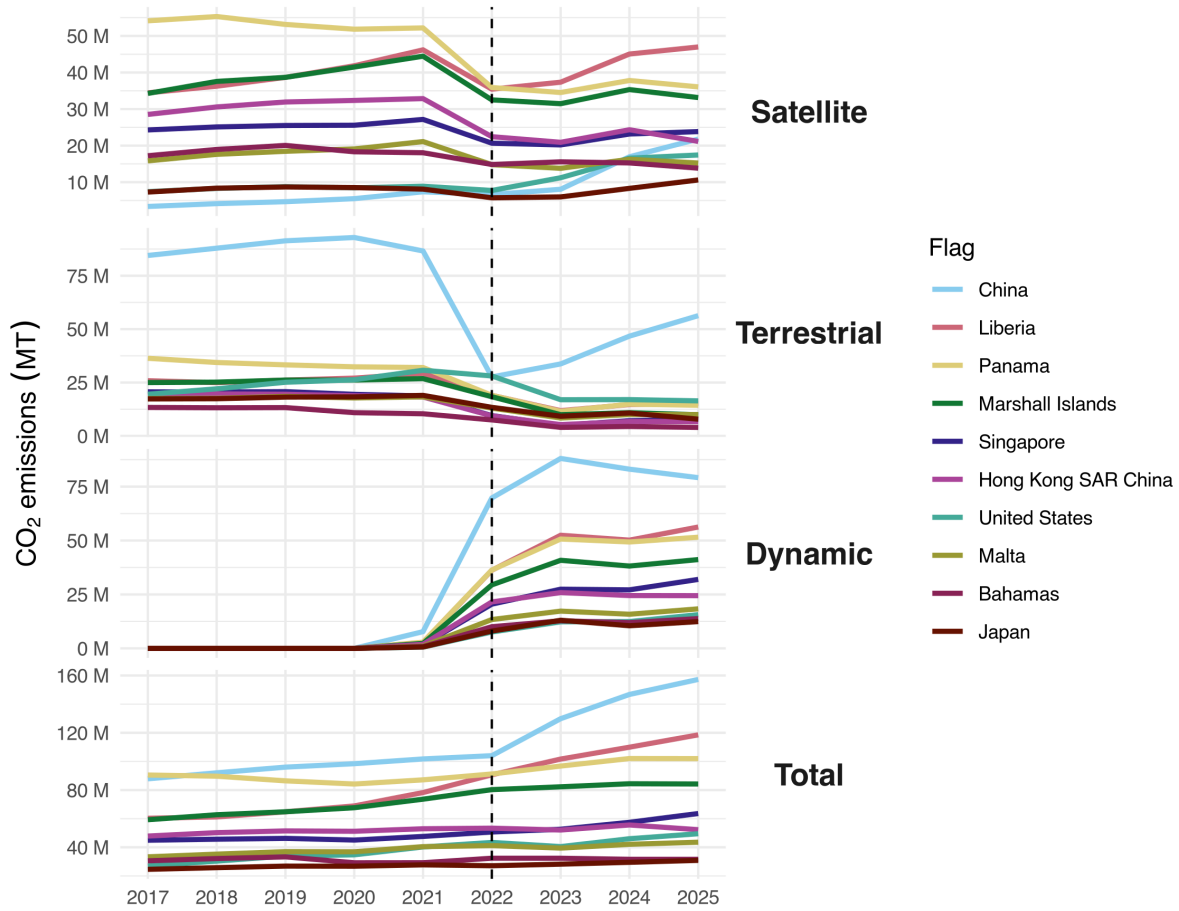


Figure 26: Annual total CO₂ emissions (metric tonnes) by AIS-broadcasting vessels, disaggregated by AIS receiver type (satellite, terrestrial, and dynamic) and total aggregated across receiver types, for the top 10 flags with the most emissions in 2025. A dashed vertical line is shown for 2022, the first year in which dynamic AIS was widely adopted.

7.6 Figure S9

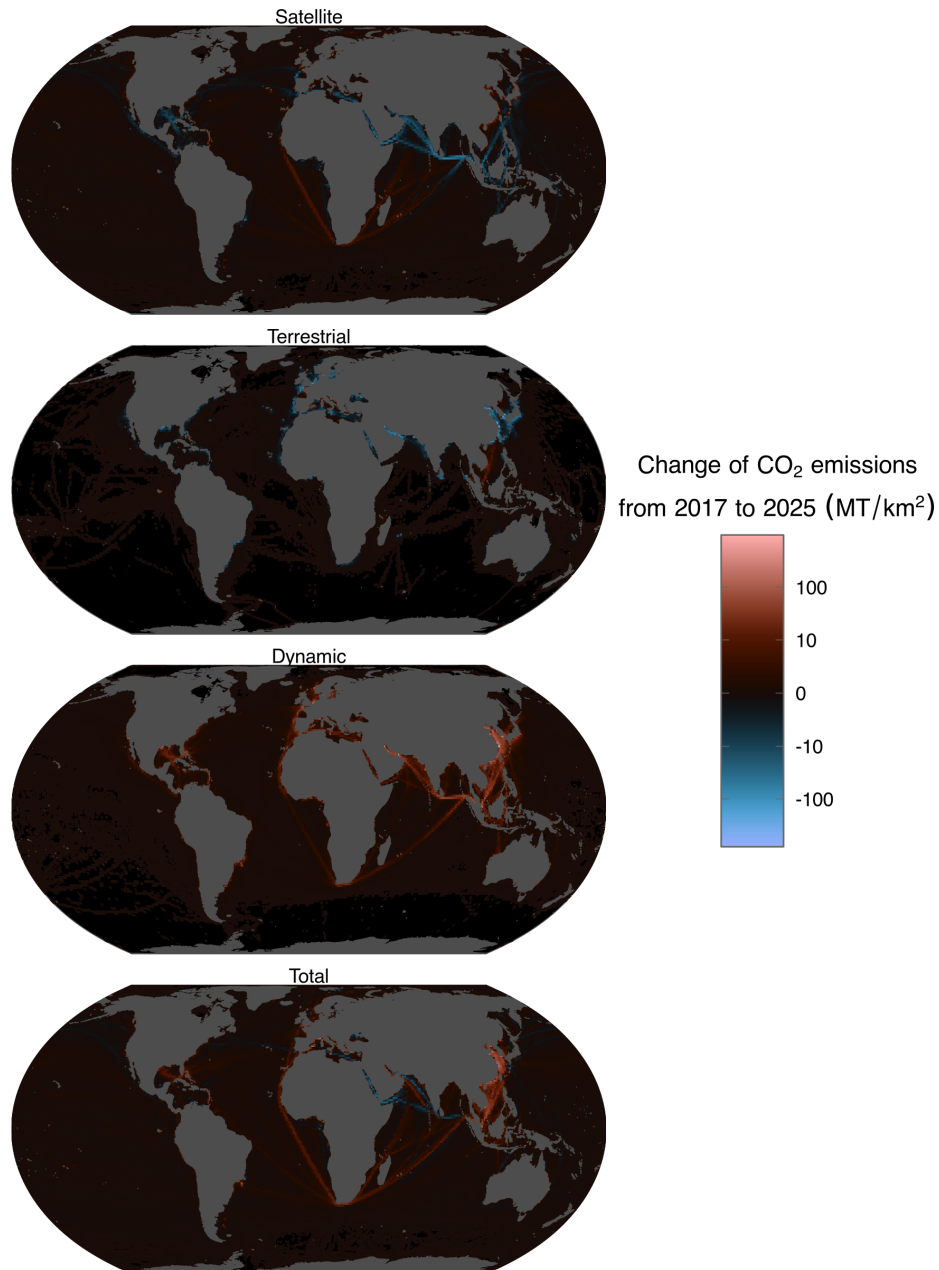


Figure 27: Spatial change in CO₂ emissions by AIS-broadcasting vessels between 2017 and 2025 (metric tonnes) by receiver type (satellite, terrestrial, and dynamic) and total across receiver types, at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

8 Inventories comparison

The figures behind the systematic comparison of our estimates against the published bottom-up inventories and the top-down EDGAR and CEDS series. The data these read is prepared by `_targets_02_inventories_comparison.R` and loaded above; the plotting happens here so every figure in the paper is drawn by this notebook.

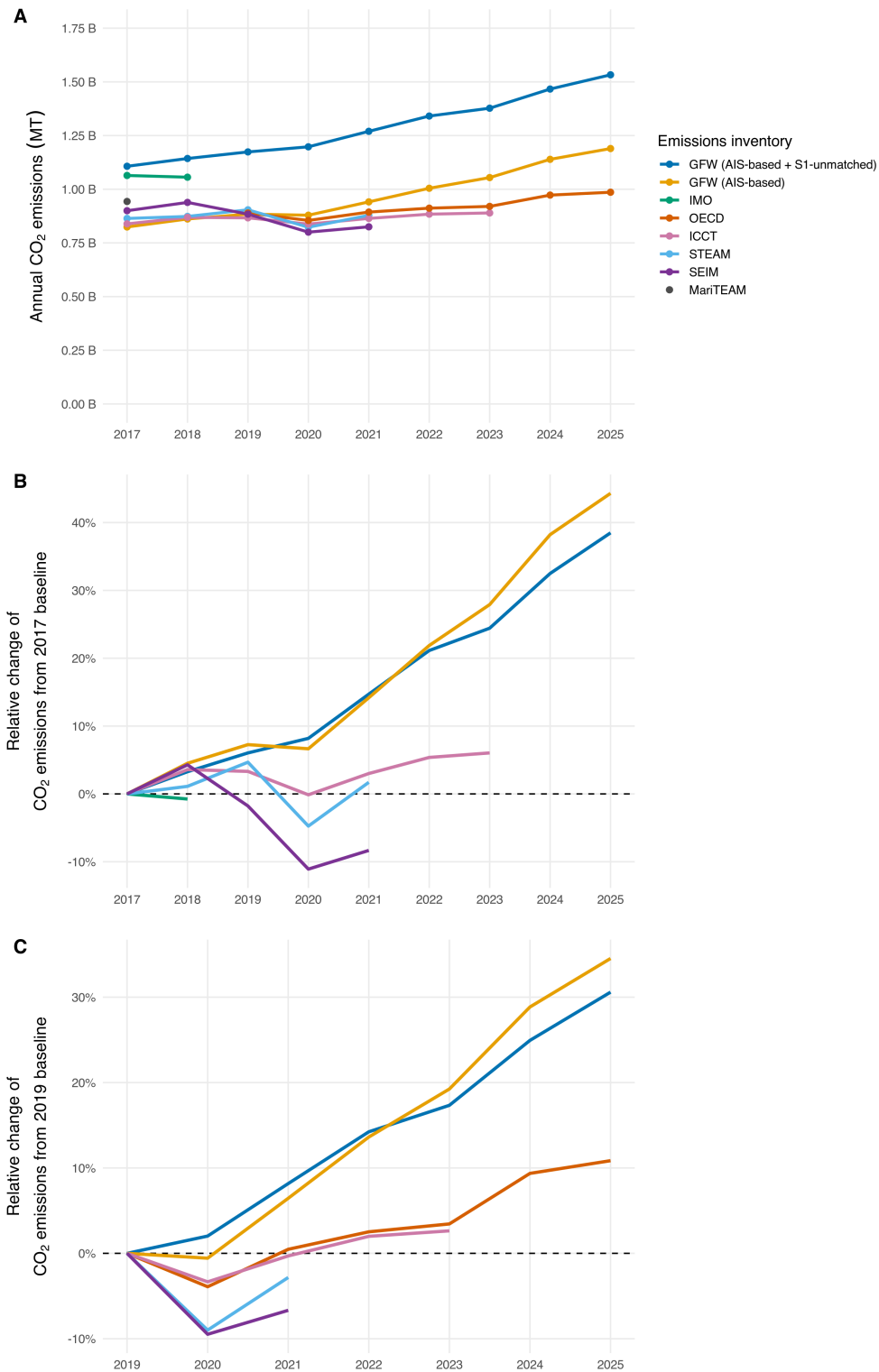


Figure 28: Our AIS and fused estimates against the published bottom-up inventories. Panel A shows levels; panels B and C show change relative to 2017 and 2019.

Table 15: Annual marine vessel CO₂ emissions (billions of metric tonnes) for our AIS-based and fused (AIS-based + S1-unmatched) estimates and for the published bottom-up, activity-based inventories. A dash marks a year for which an inventory publishes no estimate.

Year	GFW (AIS-based + S1-unmatched)	GFW (AIS-based)	IMO	OECD	SAVE (ICCT)	STEAM	SEIM	MariTEAM
2017	1.107	0.824	1.064	-	0.839	0.864	0.900	0.943
2018	1.143	0.861	1.056	-	0.869	0.874	0.939	-
2019	1.174	0.884	-	0.889	0.867	0.904	0.884	-
2020	1.198	0.879	-	0.855	0.838	0.823	0.800	-
2021	1.270	0.941	-	0.894	0.864	0.879	0.825	-
2022	1.341	1.005	-	0.912	0.884	-	-	-
2023	1.377	1.054	-	0.920	0.890	-	-	-
2024	1.467	1.139	-	0.973	-	-	-	-
2025	1.533	1.189	-	0.986	-	-	-	-

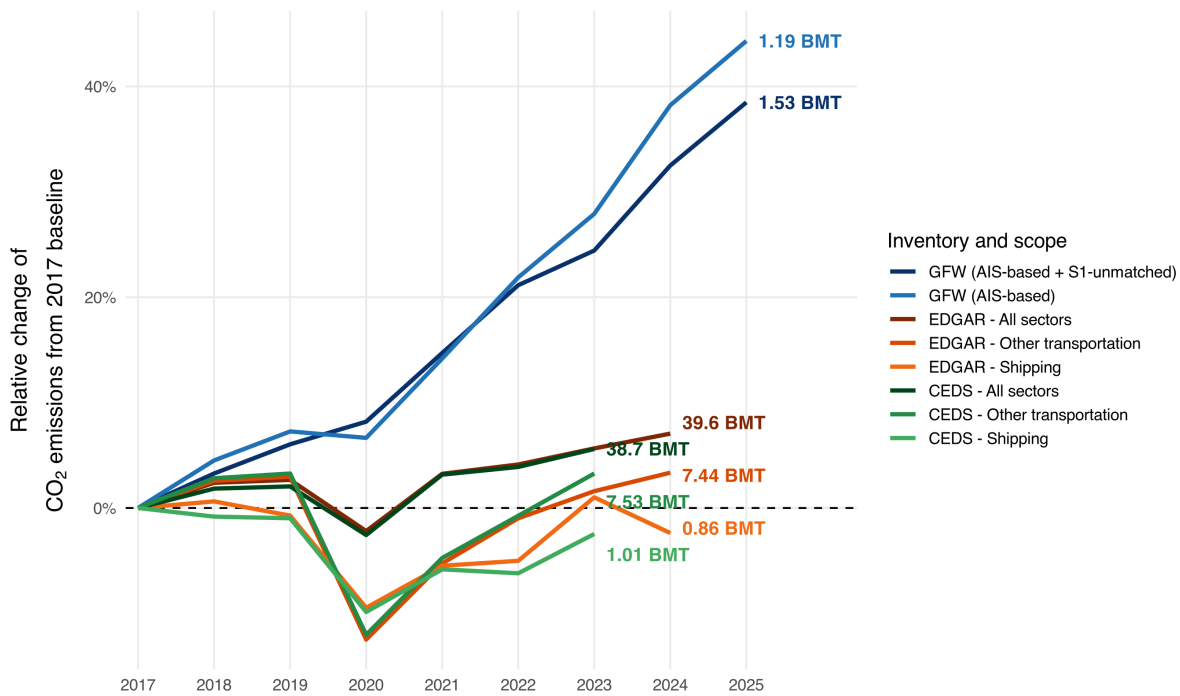


Figure 29: Marine emissions growth against all sectors and against other transport modes, in EDGAR and CEDS, indexed to 2017.

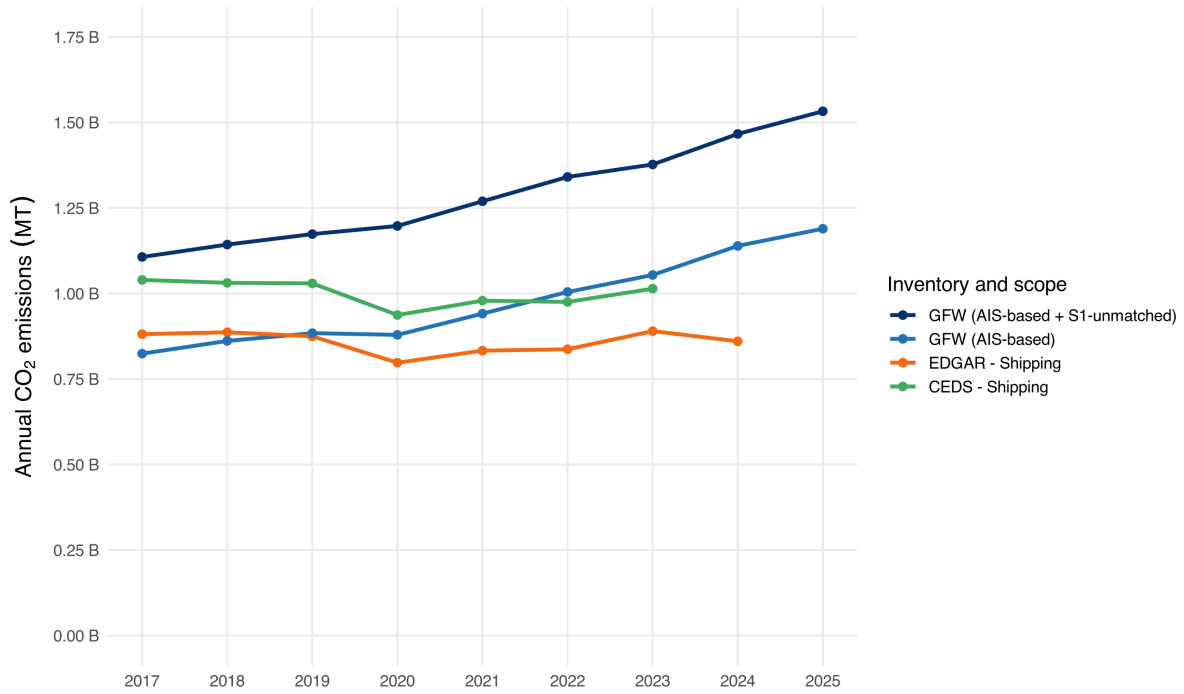


Figure 30: Our shipping totals against the shipping sector of the top-down EDGAR and CEDS inventories.

Table 16: Annual marine vessel CO₂ emissions (billions of metric tonnes) for our AIS-based and fused (AIS-based + S1-unmatched) estimates against the shipping sector of the top-down EDGAR and CEDS inventories, whose shipping scope combines international and domestic navigation. A dash marks a year for which an inventory publishes no estimate.

Year	GFW (AIS-based + S1-unmatched)	GFW (AIS-based)	EDGAR - Shipping	CEDS - Shipping
2017	1.107	0.824	0.881	1.040
2018	1.143	0.861	0.887	1.031
2019	1.174	0.884	0.875	1.030
2020	1.198	0.879	0.798	0.937
2021	1.270	0.941	0.833	0.979
2022	1.341	1.005	0.837	0.975
2023	1.377	1.054	0.890	1.014
2024	1.467	1.139	0.860	-
2025	1.533	1.189	-	-

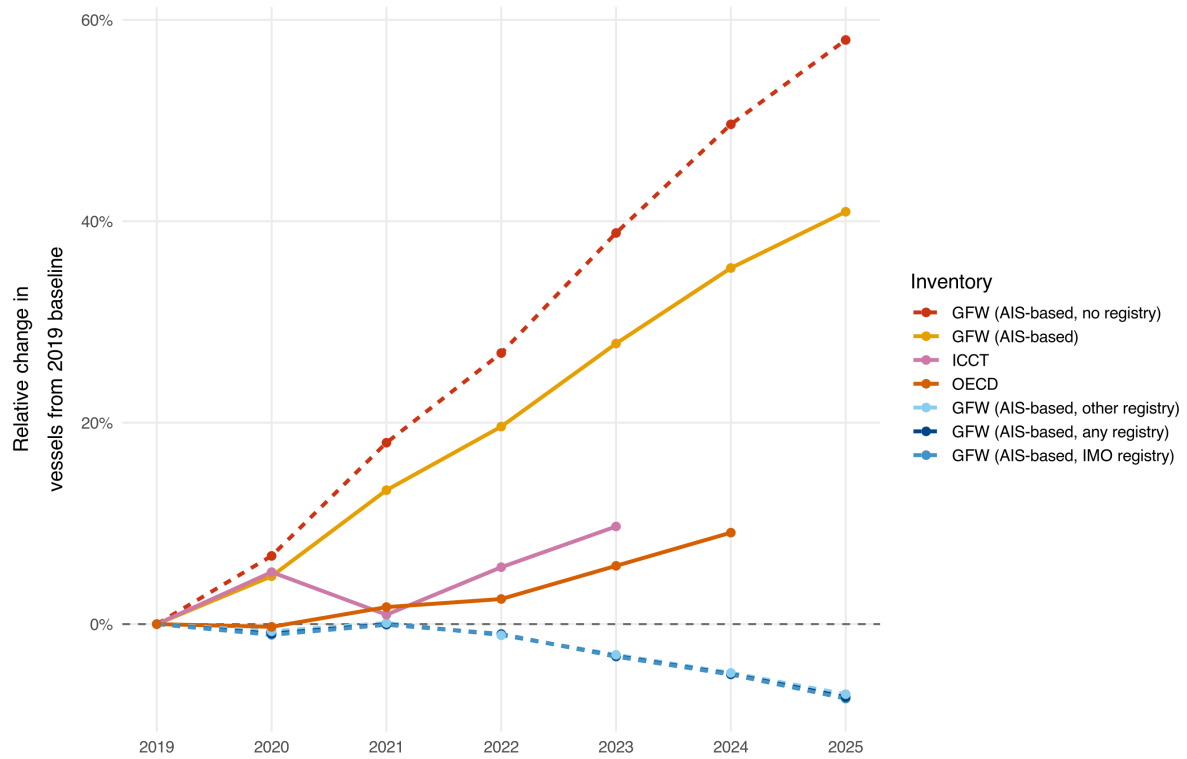


Figure 31: Modelled fleet size over time for the inventories publishing a vessel count per year, split by registry status for ours.

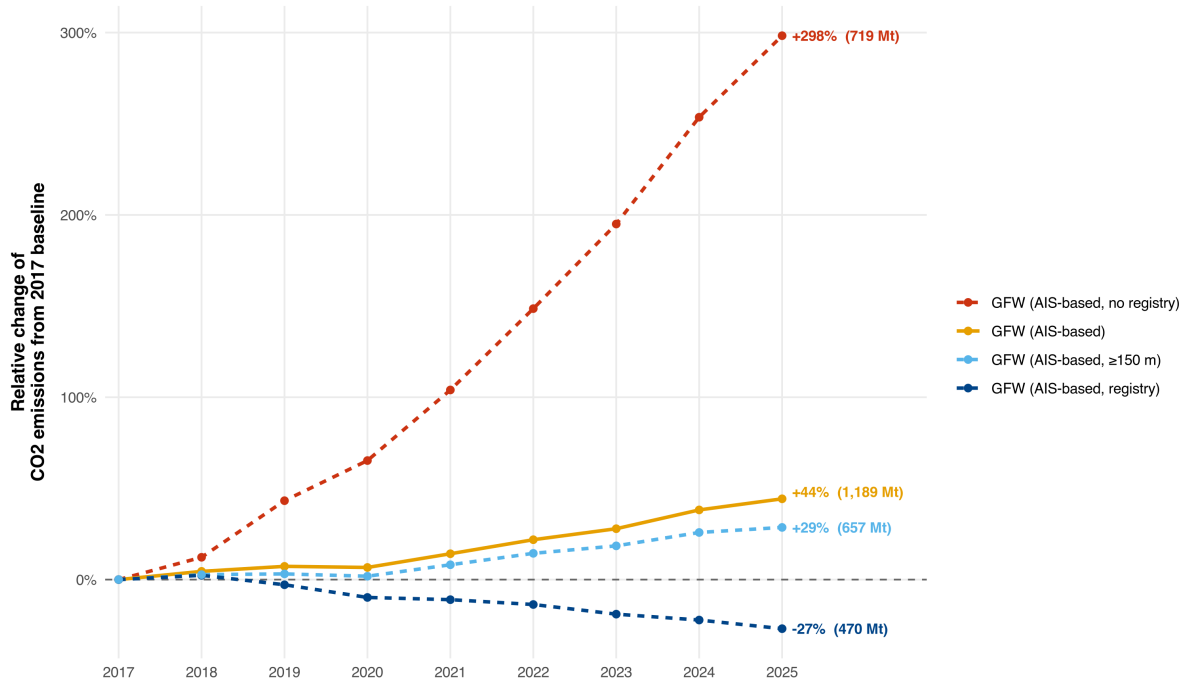


Figure 32: Emissions from registry-matched and unmatched vessels, relative to 2017, with the size-restricted fragment for comparison.

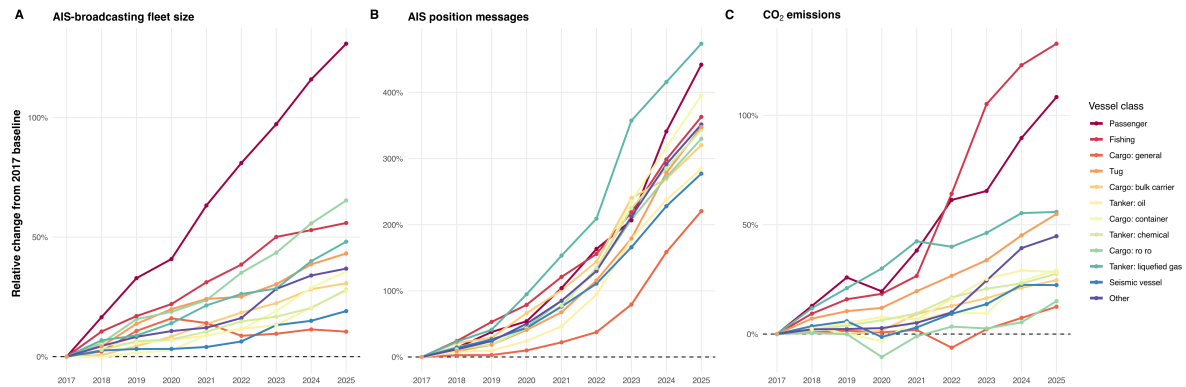


Figure 33: Growth by vessel class since 2017 in emissions, vessel count and activity.

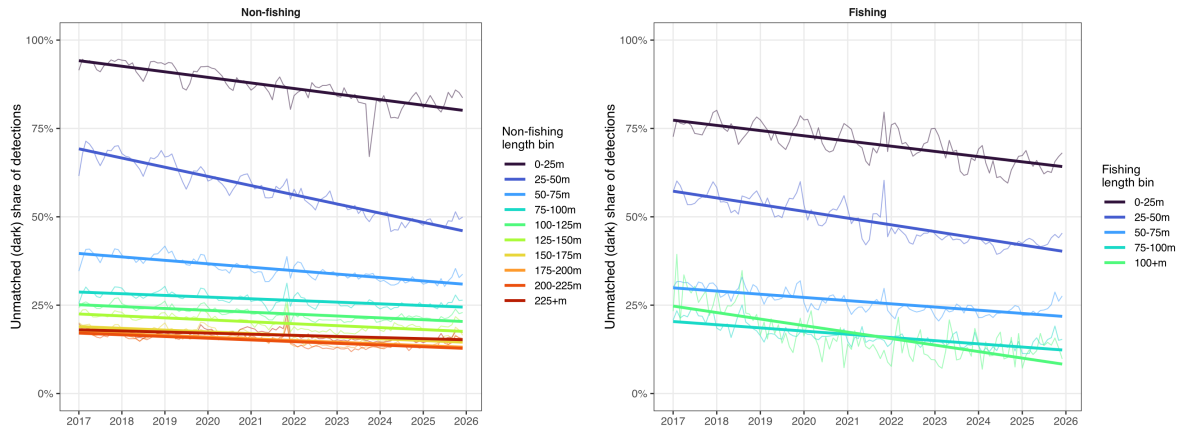


Figure 34: Share of Sentinel-1 detections with no matching AIS broadcast, by length bin, over the full period.

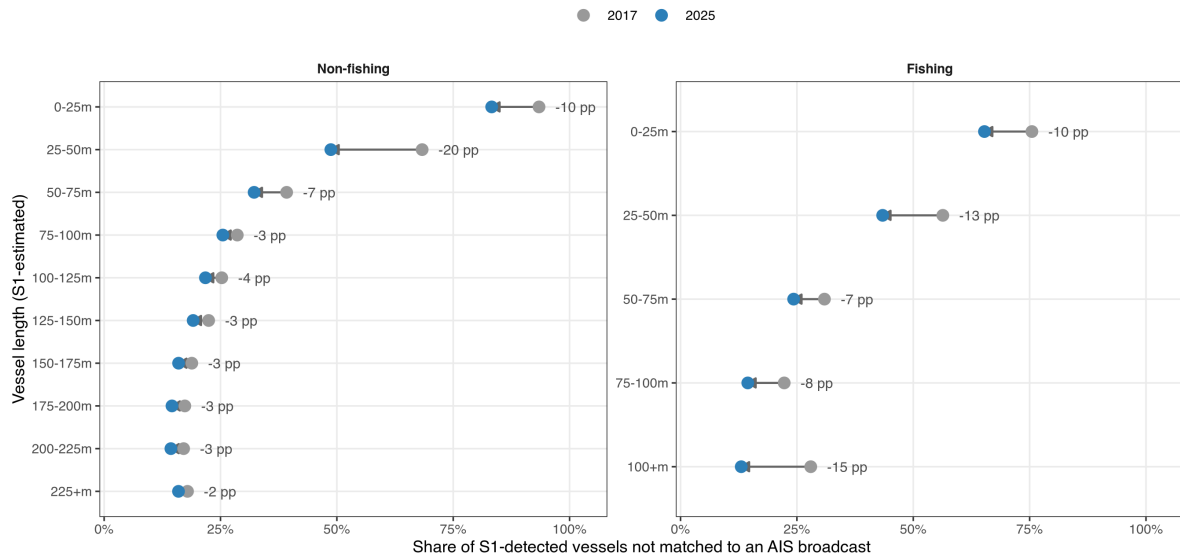


Figure 35: AIS carriage saturation by vessel size: the unmatched share is flat for the largest vessels and falls steeply for the smallest.

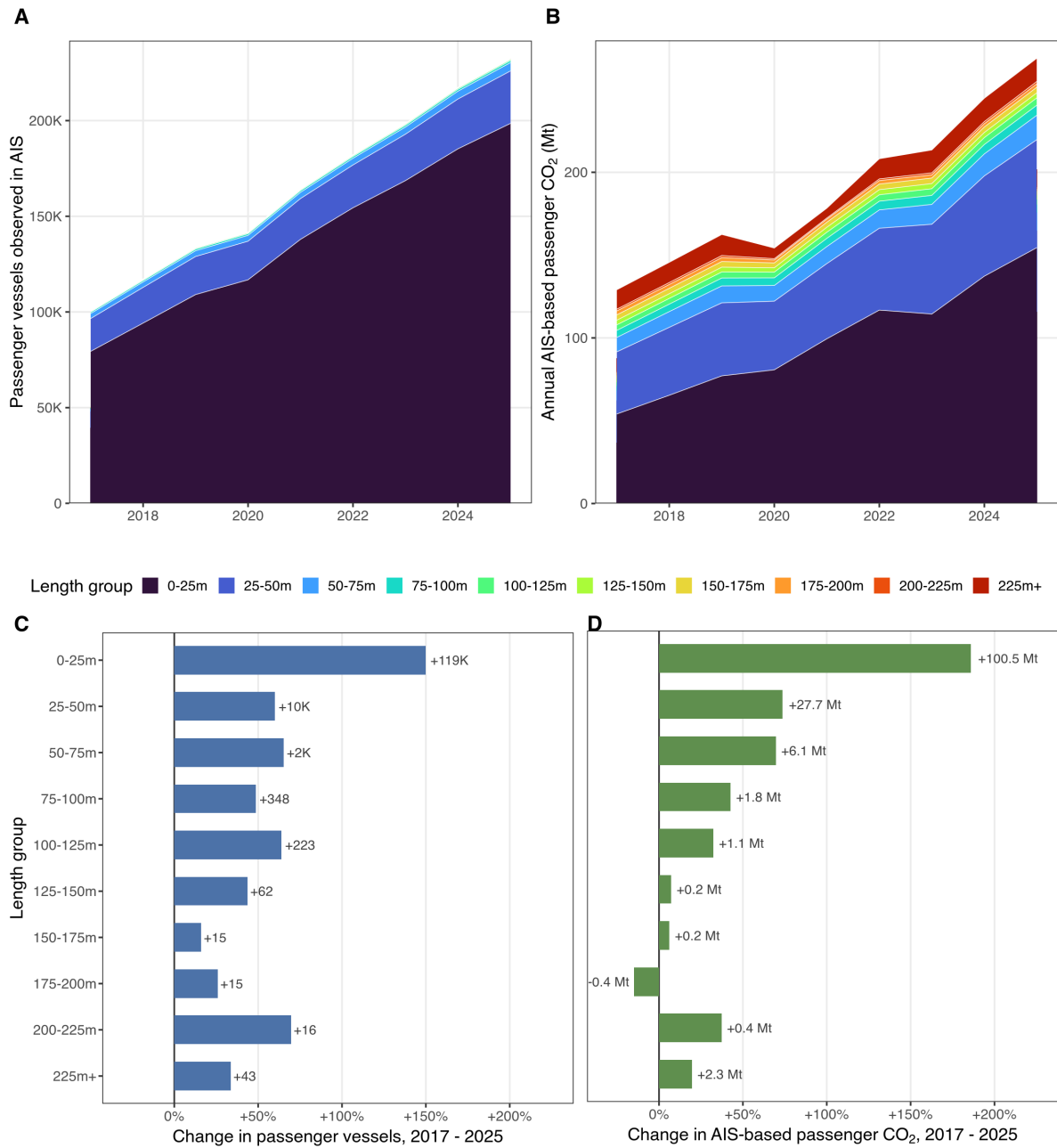


Figure 36: Passenger vessels by length bin, 2017 against 2025, showing growth concentrated in the smallest sizes.

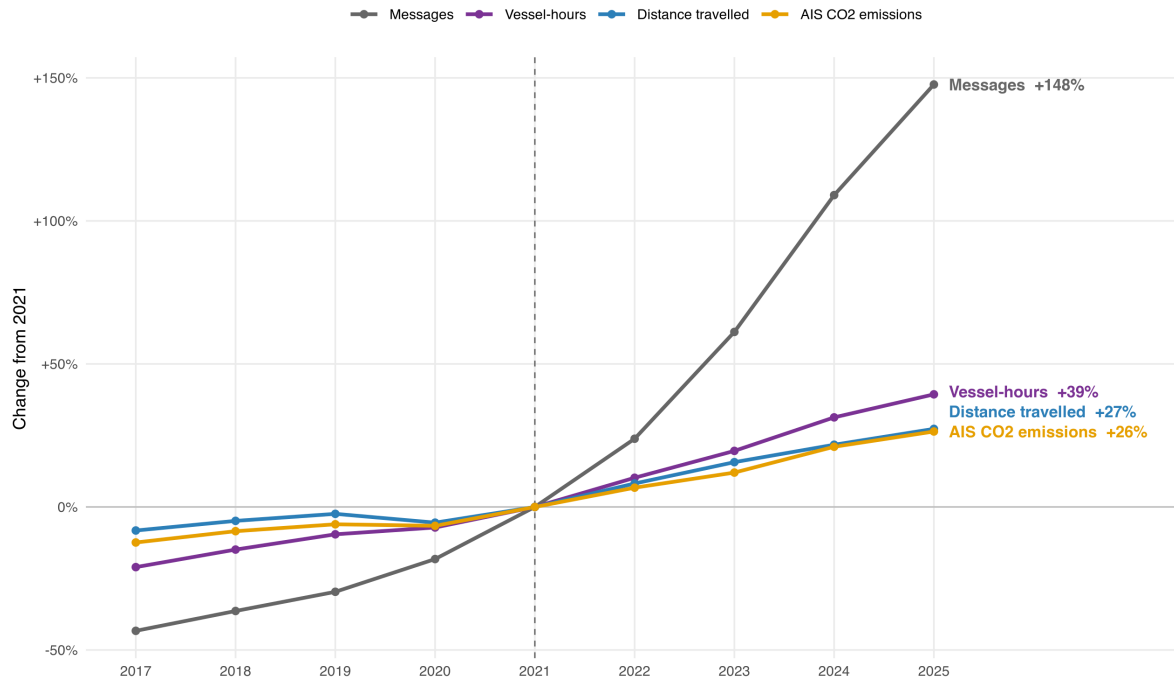


Figure 37: AIS messages received against observed activity and emissions, indexed to 2021, separating densification from new activity.

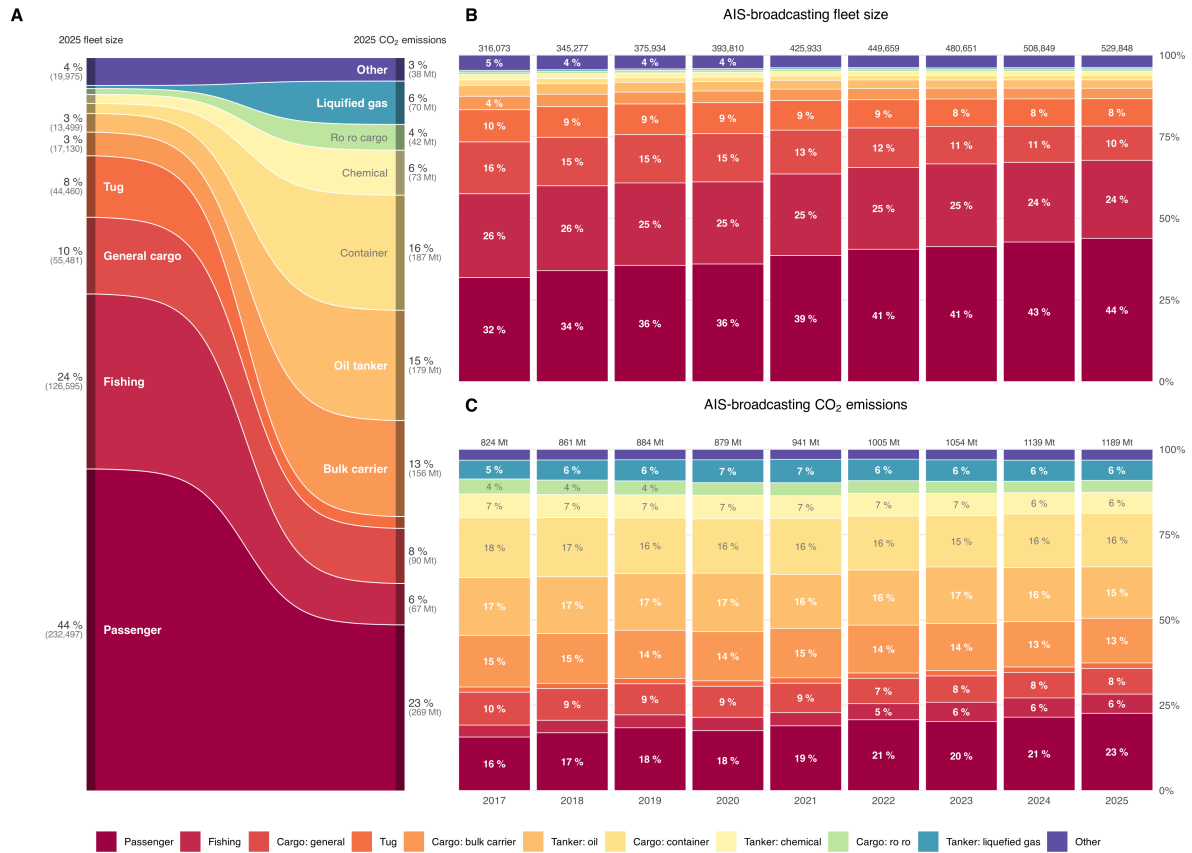


Figure 38: Fleet composition in 2025: each vessel class's share of the fleet against its share of CO₂, with the supporting series.

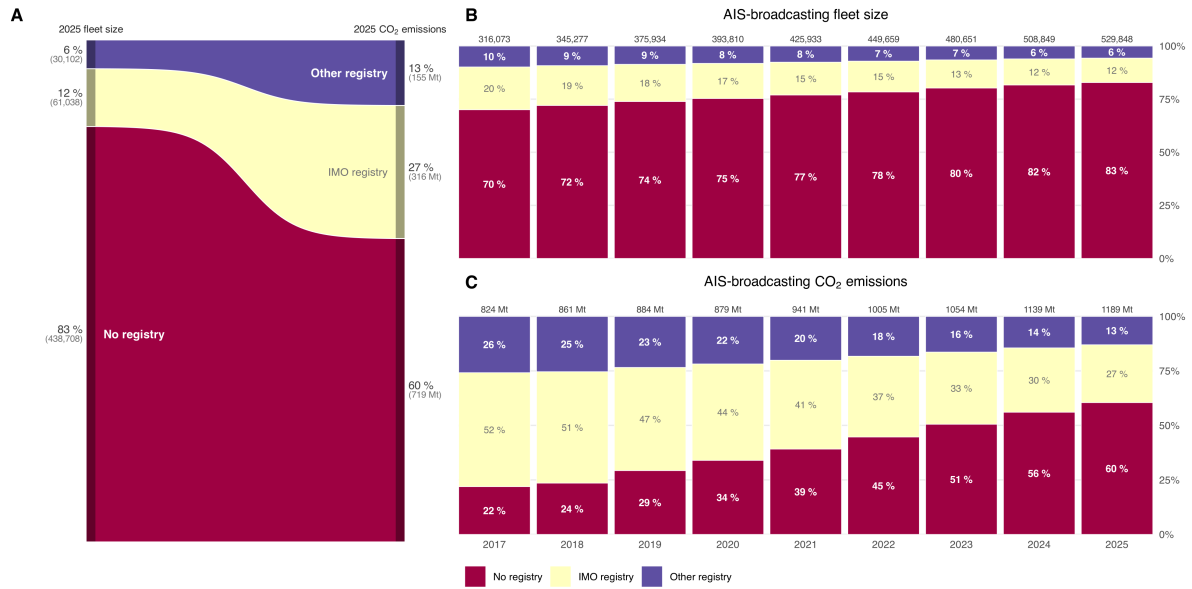


Figure 39: Registry status in 2025: share of the fleet against share of CO₂ for registry-matched and unmatched vessels, with the supporting series.

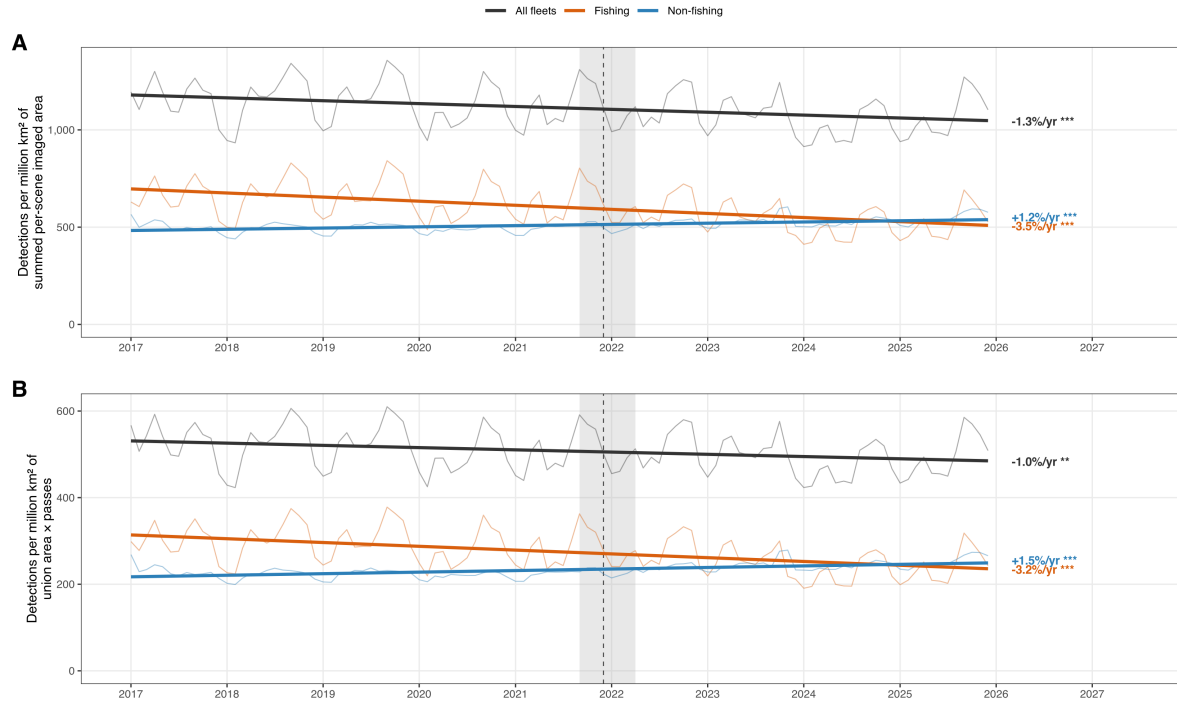


Figure 40: Sentinel-1 detection density under two denominators, and by fixed length bin, showing detected activity per unit of observing effort.

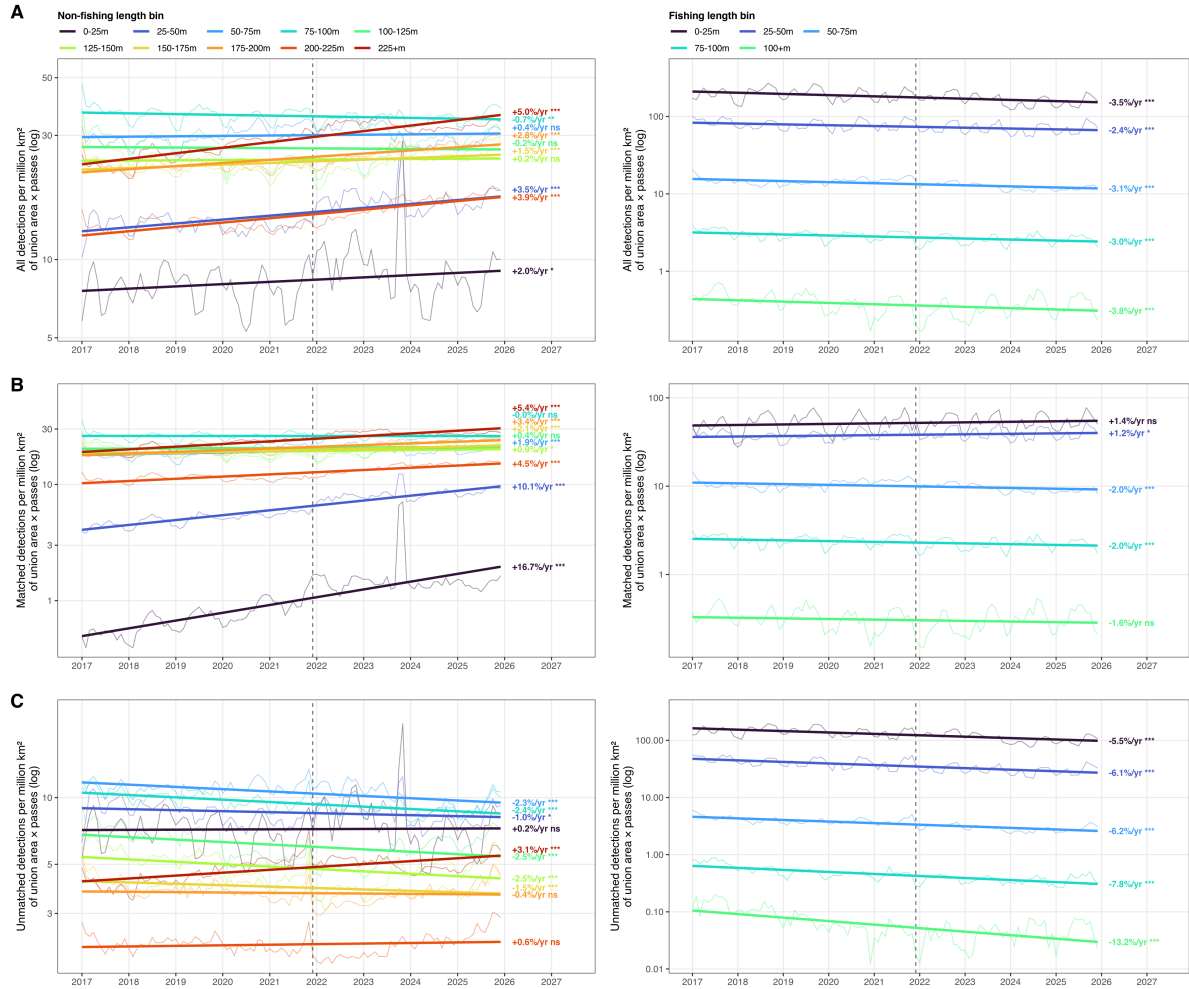


Figure 41: Sentinel-1 detection density by length bin, split into total, AIS-matched and unmatched detections, for fishing and non-fishing fleets.

8.1 Inventory intensity across models

Table 17: CO₂ intensity per vessel for each inventory in the comparison, ordered by intensity. Intensities are computed on the fleet count each model publishes: single-total models use their reported figure, per-year models the mean over years where both a count and emissions exist, and the GFW rows our own activity summary on the same basis. Per-distance intensity is shown only for the models reporting distance travelled: SAVE (ICCT) restricted to its distance-reporting classes, and our AIS series whole and registry-matched.

Inventory	CO ₂ per vessel (kt)	CO ₂ per distance (kg nm ⁻¹)
MariTEAM	20.55	—
SEIM	7.90	—
OECD	7.83	—
SAVE (ICCT distance-reporting classes)	6.78	289.5
GFW (AIS-based, any registry)	6.08	317.8
IMO	4.92	—
SAVE (ICCT)	3.39	—
GFW (AIS-based)	2.34	322.4
STEAM	2.33	—